BANK OF THAILAND

Monetary Policy Report

April 2013

Monetary Policy in Thailand

The Monetary Policy Committee

Under the Bank of Thailand Act, the Monetary Policy Committee (MPC) comprise of the Governor and two deputy Governors, as well as four distinguished external members representing various sectors of the country, with the aim to provide monetary policy decisions that are effective and transparent.

The Monetary Policy Objective

The MPC sets monetary policy with the objective to support sustainable and full potential economic growth, without causing inflationary problems or economic and financial imbalances or bubbles.

The Monetary Policy Target

The Cabinet approves the use of quarterly average core inflation with the range of 0.5-3.0 percent as its policy target for 2013, as proposed by the MPC together with the Minister of Finance. In the event that the target is missed, the MPC is required to explain the reasons and resolution thereof to the public.

The Monetary Policy Instrument

The MPC utilizes the 1-day bilateral repurchase transaction rate as the key policy rate to signal the monetary policy stance.

Evaluating economic conditions and forecasts

The Bank of Thailand considers information from all sources, the macroeconomic model, data in each economic sector, as well as direct opinions from large enterprises and small and medium-sized enterprises from all over the country, and various financial institutions to ensure that economic evaluations and forecasts are accurate and covers all aspects, both in the macro and micro levels.

Monetary Policy Communication

Recognizing the importance of Monetary Policy communication to the public, the MPC arrange many forms of communication, both in Thai and English, such as (1) organizing a press statement at 14.00 hrs. on the day of the meeting, (2) publishing Minutes of the MPC meeting two weeks after the meeting, and (3) publishing the Monetary Policy Report every quarter.

Monetary Policy Report

The *Monetary Policy Report* is prepared quarterly by staff of the Bank of Thailand with the approval of the Monetary Policy Committee (MPC). It serves two purposes: (1) to communicate to the public the MPC's consideration and rationales for the conduct of monetary policy, and (2) to present the latest set of economic and inflation forecasts, based on which the monetary policy decisions were made.

The Monetary Policy Committee
April 2013

Mr. Prasarn Trairatvorakul	Chairman
Mrs. Pongpen Ruengvirayudh	Vice Chairman
Mrs. Tongurai Limpiti	Member
Mr. Ampon Kittiampon	Member
Mr. Narongchai Akrasanee	Member
Mr. Siri Ganjarerndee	Member
Mr. Aswin Kongsiri	Member

Monetary Policy Report

April 2013

Contents

1. Growth and Inflation Prospects and Monetary Policy 1

1.1 Growth and inflation prospects 1

1.2 Economic outlook 4

1.3 Monetary policy decisions 19

1.4 Appendix 21

BOX: What are the determinants of capital inflows to Thailand? 24

2. Recent Economic Developments 27

2.1 The global economy 27

2.2 The domestic economy 31

2.3 Costs and prices 40

BOX: Budget sequestration and implications on the US economy 43

BOX: Appreciation of the Thai baht and its implication on exports 45

3. Monetary and Financial Stability 49

3.1 Financial markets 49

3.2 Financial institutions 52

3.3 Non-financial sectors 55

Growth and Inflation Prospects and Monetary Policy

1. Growth and Inflation Prospects and Monetary Policy

The Thai economy was expected to experience continuous growth throughout the projection period. Domestic demand would still benefit from private consumption and investment momentum spurred by various supportive factors. Meanwhile, public spending was expected to have an increasing role in economic growth, while exports were likely to recover gradually in line with global economic conditions. Inflationary pressure was poised to pick up gradually going forward, consistent with improvements in the domestic economy and higher-than-expected global oil prices.

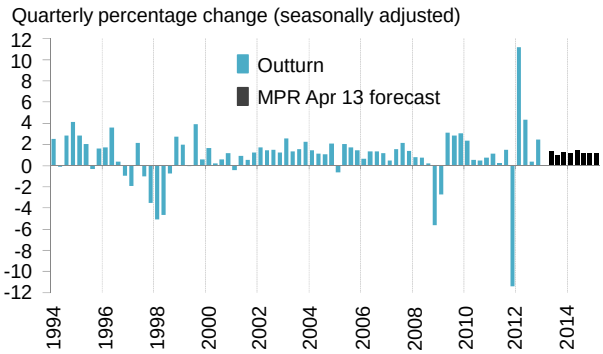
Given the fragile state of the global economy and domestic financial stability risks, the MPC judged that an accommodative monetary policy stance remained appropriate. The MPC thus voted to hold the policy rate at 2.75 percent per annum in its last two meetings.

1.1 Growth and inflation prospects

The Thai economy was likely to experience continuous growth, while inflation pressure was poised to pick up.

The Thai economy was expected to expand continuously with private demand remaining the main engine of growth, backed by strong supporting factors including accommodative monetary and fiscal conditions. In 2013, the MPC assessed that overall private demand would moderate towards the normal trend, after having accelerated sharply in the previous year. Although private demand was likely to moderate slightly from previously assessed, higher public stimulus and gradual recovery in exports following trading partners' economic recovery would help support overall

Chart 1.1 Thailand's GDP growth



Source: Office of the National Economic and Social Development Board and calculation by Bank of Thailand
Note: At 1988 prices (seasonally adjusted)

domestic demand momentum to continue into the next periods. Given that underlying economic factors and major economic assumptions had not changed much, the MPC assessed that the overall Thai economic momentum would likely remain similar to that previously expected. Nevertheless, a slight upward adjustment on economic growth projection was made to reflect the higher-than-expected growth in 2012 Q4, which raised the forecast base, as well as higher public stimulus **(Chart 1.1 and Table 1.1)**.

Table 1.1 Forecast summary

Percent	2012*	2013	2014
GDP growth	6.4	5.1	5.0
	(5.9)	(4.9)	(4.8)
Headline inflation	3.0	2.7	2.7
	(3.0)	(2.8)	(2.6)
Core inflation	2.1	1.6	1.7
	(2.1)	(1.7)	(1.6)

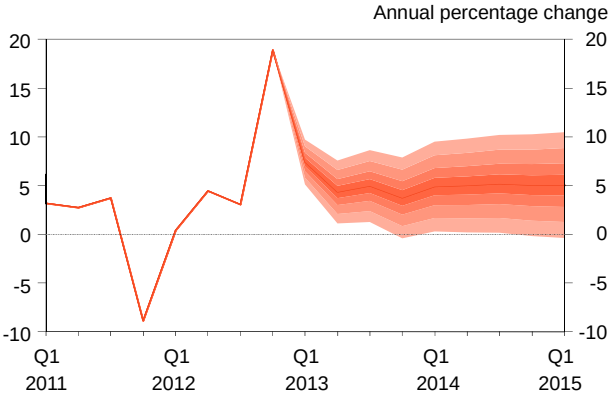
Note: * Outturn
( ) Monetary Policy Report January 2013
Source : Office of the National Economic and Social Development Board and calculation by Bank of Thailand

Fragile recovery of the global economy, in particular the euro area and the US, remained the main risk that could cause the Thai economy to expand less than the baseline projection. Nevertheless, the MPC assessed that this risk did not change significantly from the previous assessment due to progress in the resolution of the euro area's sovereign debt problems and US fiscal problems. Meanwhile, the situations in Italy and Cyprus were expected to be a short-term risk factor and should not develop into a systemic risk in the financial sector going forward. **At the same time, upside risks from domestic factors increased** from greater clarity regarding the 2 trillion baht investment in infrastructure, which could be disbursed at a faster

pace than assessed in the baseline case. Overall, the MPC assessed that risks to economic growth in future periods would become balanced as opposed to that skewed downwards in the previous projection. The fan chart for growth was thus balanced throughout the projection period **(Chart 1.2)**.

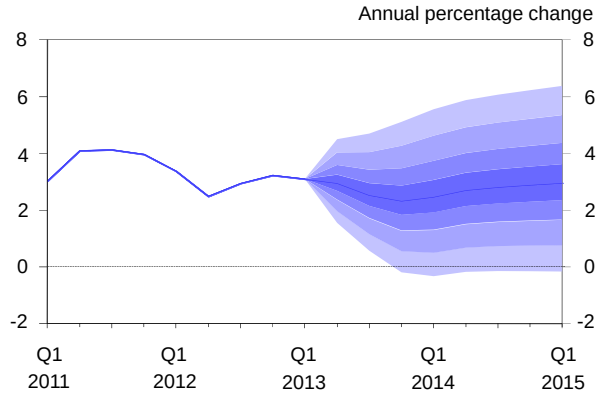
Inflation was likely to remain stable. Inflationary pressure in the short-term could be somewhat lower than expected, as reflected by the inflation outcome in 2013 Q1 which was lower than assessed, particularly raw food prices. However, going forward, the MPC deemed that demand pressure would likely increase gradually in line with the domestic economic trend. Meanwhile, cost pressure was likely to pick up slightly from the previous assessment, due to higher global oil prices than previously assumed. As a result, the baseline projection for inflation remained relatively unchanged **(Table 1.1)**. Going forward, the MPC assessed that risks to inflation would slightly tilt towards the upside. Inflationary pressure from shortage of labor and other production factors which could pass through to prices at a greater extent during economic expansion would warrant monitoring. Fan charts for inflation were thus skewed upwards slightly, compared to those that were previously balanced **(Charts 1.3 and 1.4)**.

Chart 1.2 GDP growth forecast



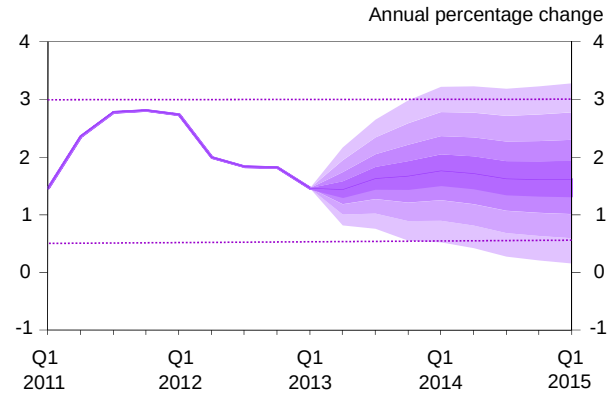
Note: The fan chart covers 90 percent of the probability distribution.

Chart 1.3 Headline inflation forecast



Note: The fan chart covers 90 percent of the probability distribution.

Chart 1.4 Core inflation forecast



Note: The fan chart covers 90 percent of the probability distribution.

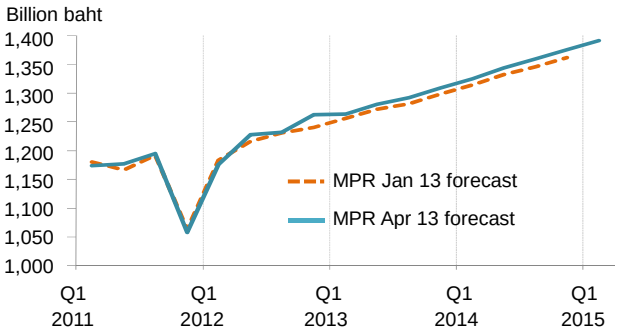
1.2 Economic outlook

Momentum from robust domestic demand should continue, while the export sector would recover gradually in tandem with improving global economic outlook.

The Thai economy was expected to expand continuously.

The Thai economy was likely to grow continuously, fueled mainly by private demand thanks to strong supporting factors including accommodative fiscal and monetary conditions. In 2013, private demand would moderate toward its normal trend after having accelerated sharply in the previous periods. Momentum during the beginning of the year could weaken somewhat from the previous assessment. Private consumption was poised to slow down slightly due to discontinued acceleration in automobile spending despite strong consumption growth in other categories. Business sector investments were likely to be more gradual, albeit still on an up-cycle trend. In the meantime, the MPC assessed that fiscal stimulus would be higher than last projected and would play a larger role in economic growth in 2013 than previously assessed. This was due to the slight increase in budget spending and the incorporation of the 2 trillion baht infrastructure investment project into the baseline case. The increase in fiscal stimulus would help sustain domestic demand growth going forward **(Chart 1.5 and Table 1.2)**.

Chart 1.5 Level of GDP



Source: Office of the National Economic and Social Development Board and calculation by Bank of Thailand
Note: At 1988 prices (seasonally adjusted)

Table 1.2 Forecasts for GDP and components		
Percent	2013	2014
GDP growth	5.1	5.0
Domestic demand	6.2	5.2
Private consumption	4.7	3.7
Private investment	7.4	9.4
Government consumption	4.5	4.1
Public investment	22.3	6.9
Exports of goods and services	7.2	9.3
Imports of goods and services	9.2	10.3

Note: At 1988 prices

With regard to the external sector, the MPC assessed that merchandise exports were likely to recover slowly, in line with global economic recovery. Exports were expected to play a larger role as engine of growth in late 2013 and support economic growth momentum when government stimulus measures tapered off. At the same time, the MPC assessed that imports would likely soften from the previous projection, in line with lower-than-expected investment growth prospects of the business sector. Meanwhile, overall exports were likely to remain similar to the previous assessment, in part from anticipated high growth in the tourism sector. As a result, net exports would lend greater support to economic growth than previously assessed.

Given insignificant changes in economic factors and major economic assumptions, the MPC assessed that momentum of the Thai economy would be similar to that assessed previously (Chart 1.5). However, economic outturns for 2012 Q4, especially private consumption and public spending, were much higher than expected, raising the forecast base and slightly lifted up growth forecast in 2013. Nevertheless, this upside effect was reduced by a weaker-than-expected outlook of

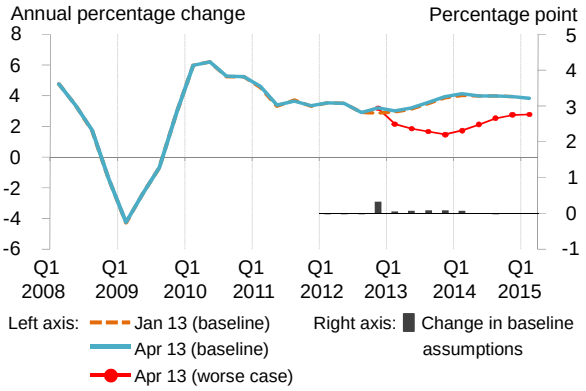
private demand in the beginning of the year. Economic projections for 2013 were thus adjusted up only slightly. For 2014, economic projections were also adjusted upwards from last time, partly due to the anticipated increase in fiscal spending which would help shore up domestic demand growth going forward.

The global economy to recover gradually

As expected, the global economy recovered at a slow pace, with some downside risks in the short run.

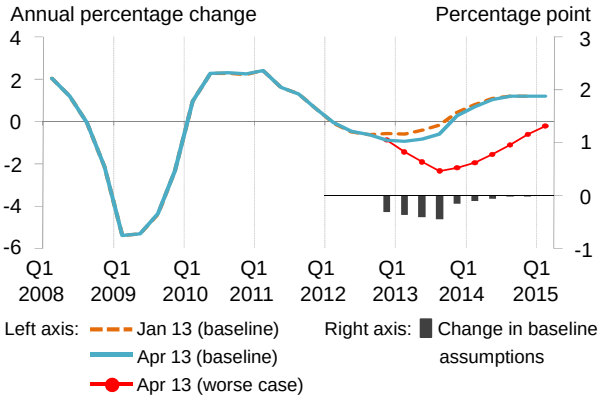
Thailand's trading partners' economies were expected to recover slowly throughout the projection period (Chart 1.6), with a possible temporary slowdown in early 2013. This was partly due to weaker momentum from the US and euro area economies continuing from 2012 Q4 which exhibited lower-than-expected economic growth. However, for the rest of the projection period, the MPC assessed that Thailand's trading partners' economies would recover slightly faster than previously expected, thanks to improvements in Japan's economic outlook from more concrete stimulus measures, gradual recovery of the US economy and continued expansion of the Asian economies.

Chart 1.6 Growth assumptions for Thailand's trading partners



Note: Weighted by each country's share in Thailand's total exports

Chart 1.7 Growth assumptions for the euro area



The outlook for euro area economies weakened from the previous assessment (Chart 1.7). Economic momentum in the first half of 2013 was expected to be softer than previously assessed. This was due to the greater-than-expected economic contraction in 2012 Q4 of the euro area, as a result of weaker exports. In addition, uncertainties of the situations in Italy and Cyprus during the start of the year could somewhat affect economic activities in the short run. Overall, however, euro area economies were expected to return to the previously projected recovery path in late 2013, led by the recovery of the export and manufacturing sectors, especially in

Germany which began to show signs of economic improvement. Meanwhile, fiscal austerity measures and the tightening credit conditions by financial institutions would continue to constrain domestic demand recovery going forward. Risks to the euro area in the next periods did not change significantly as there was some progress in resolving sovereign debt problems and the situations in Italy and Cyprus were expected to be only temporary and would not develop into a systemic risk in the financial sector. The euro area's economic growth assumption in the worse case scenario thus remained nearly unchanged.

At the same time, the US economy was likely to improve gradually going forward (Chart 1.8). US economic momentum in 2013 was lower than previously projected, due to weaker-than-expected economic conditions in 2012 Q4. However, looking ahead, the MPC assessed that the US economy would be supported by the likely increase in private consumption and investment, as well as gradual improvements in labor market conditions. This would help offset most of the impact from sequestration that will come into effect in March 2013. Nevertheless, fiscal problems would continue to hamper recovery going forward, particularly from uncertainties regarding the debt ceiling. There is a possibility that the US government may consider compensating the debt ceiling increase with a further reduction in fiscal deficit, which could affect confidence and economic activities of the private sector.

Japan's economy showed signs of stability with a promising outlook going forward (Chart 1.9). The new government's stimulus measures became more concrete, both the supplementary budget of around 2 percent of GDP and the continuous accommodative monetary policy stance to help pull the Japanese economy out of deflation. The

Chart 1.8 Growth assumptions for the US

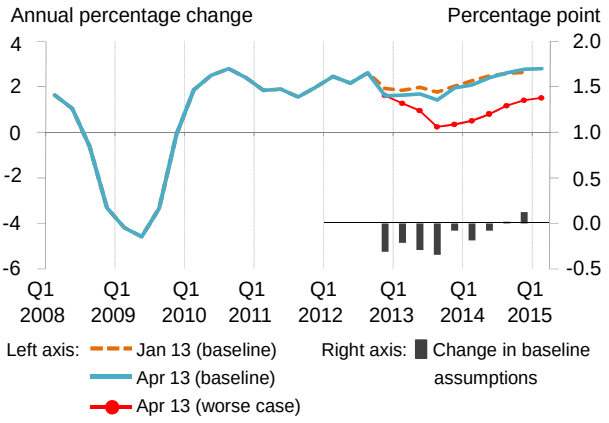
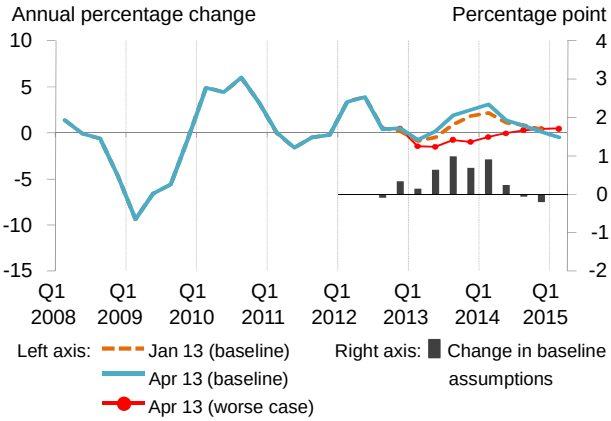
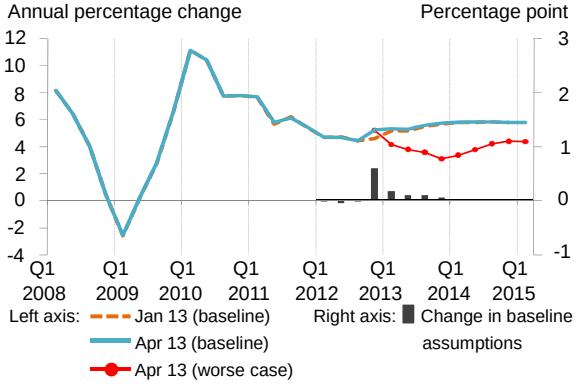


Chart 1.9 Growth assumptions for Japan



MPC assessed that impacts from the measures would gradually pass through to the real economy in the next periods, which would help support economic recovery in Japan. Meanwhile, the export sector would gain support from global demand recovery and the weaker Japanese yen.

Chart 1.10 Growth assumptions for Asian economies (excluding Japan)



Note: Weighted by each country's share in Thailand's total exports

In addition to the likely pickup in the Japanese economy, the Chinese and other Asian economies were also poised to grow strongly (Chart 1.10) on the back of robust private demand supported by accommodative fiscal and monetary conditions. Furthermore, exports of the regional economies started to show signs of improvement, particularly in electronics. On the other hand, signs of financial stability risks in the real estate sector and lending in some countries became more apparent. As a result, many economies started to implement measures to maintain economic stability during this period.

The Thai export sector picked up gradually, in line with the global economy.

The Thai export sector started to stabilize and would recover gradually towards the end of 2013. Meanwhile, imports decelerated in line with a gradual pace of private investment.

The export sector was still affected by the weak global economy. However, it started to show signs of stabilization in the beginning of 2013 and was likely to pick up steadily (Table 1.3). Latest data from the first two months of the year exhibited a slowdown in merchandise exports, particularly in automobiles, electronics, as well as agricultural and fishery products. This could lead to lower merchandise export in 2013 Q1 than previously assessed, but should recover close to the previous projection trend in late 2013. The recovery would be in line with (1) gradual recovery of trading partners' economies

and lower global economic uncertainties and (2) improvements in key leading indicators, such as the manufacturing production index, new export orders, and the semiconductor book-to-bill ratio. However, the MPC assessed that export growth in the future periods may not be as high as in the past because major economies were likely to expand below normal trends for some time. This would cause some exporters to turn more to the domestic market. Furthermore, due to domestic labor shortage problems, more businesses moved their production base to neighbor countries.

Table 1.3 Forecasts for the external sector

	2013	2014
Growth in value of exports* (F.O.B., percent)	7.5	10.0
Growth in value of imports* (F.O.B., percent)	8.7	10.3
Trade balance (billion U.S. dollars)*	6.2	6.0
Current account balance (billion U.S. dollars)*	1.8	0.1

Note: *Data revision according to definitions in IMF's Balance of Payments Manual, 6th edition (BPM6), and Ministry of Commerce's revised database

At the same time, merchandise imports in 2013 were expected to expand continuously (Table 1.3). Imports of consumer goods likely increased in line with strong demand for consumption and robust income outlook. Imports of capital goods would benefit from industrial upgrading trend to replace manual labor and the government's large-scale investment projects. However, compared to the previous projection, the MPC appraised that merchandise imports would slow down slightly on the back of lower-than-expected reconstruction in early 2013 and a slow increase in capital goods imports which were likely to stretch over the medium to long term.

With regard to currency risks, the MPC evaluated that it could cause some impact but would not hamper overall export momentum. The strong baht which strengthened faster than regional currencies during the beginning of the year

somewhat affected export revenue in baht terms. However, the impact was unlikely to be significant because some exporters had expected the baht appreciation and had hedged against it. Meanwhile, the weaker Japanese yen was also unlikely to affect Thai exports on the whole as the resulting improvement in Japanese exports would in turn help boost Thai exports of intermediate goods and parts in the supply chain. Furthermore, the Thai economy would benefit from cheaper imports of machineries from Japan, to enhance production efficiency going forward.

The MPC assessed that the current account balance[1] was likely to show a lower surplus going forward (Table 1.3) owing to higher imports in line with recovery of the export sector, business investments and public investments in large-scale infrastructure projects. Nevertheless, the current account projection for 2013 increased from the previous projection due mainly to expansions in the tourism sector.

Momentum from private demand would continue.

Private consumption and investment were expected to expand continuously, although slowing down towards the normal trend after having accelerated strongly in the previous periods.

Private consumption was likely to expand continuously and would be the main engine of economic growth in Thailand throughout the projection period. The MPC assessed that, with existing supporting factors and a larger role of public spending, private consumption in 2013 would expand more than previously projected, although the actual economic outturns indicated that consumption of durable goods had already accelerated sharply during the end of last year and there were signs of

[1] Including reinvested earnings

deceleration during the start of this year. Factors that continued to help bolster the momentum of private consumption included (1) improved purchasing power following favorable income prospects owing to the increase in minimum wage and civil servant salary, restructuring of the personal income tax rate, as well as a likely increase in farm income due to the rising trend of the prices of key agricultural products, such as rubber, palm oil and cassava following the increase in global demand, while price of rice would still be supported from the government's rice pledging scheme; (2) accommodative monetary policy which supported household credit growth; and (3) healthy consumer confidence stemming from government stimulus measures and a stable inflation trend.

The MPC expected that overall consumption would decelerate to its normal trend. Spending on consumption may not accelerate as before, partly due to the discontinuation of some public measures, front-loaded acceleration of durable goods consumption in the previous periods and debt burden of households which could cause consumers to become more money conscious.

Private investment should expand further, with its up-cycle trend to continue. In 2013, investments were expected to grow close to the long-term average, owing to investments by businesses to accommodate high demand for goods and upgrading and restructuring of its manufacturing production process to replace labor shortage. However, the MPC assessed that the business sector would likely invest at a gradual pace stretching over the medium to long term which could cause investments to moderate towards the normal trend faster than expected. The MPC appraised that many supportive factors remained strong, in particular (1) businesses remained confident in the Thai economy and were likely to invest according to their long-term plans, as reflected by high levels of

application and issuance of investment promotion certificates by the Board of Investment (BOI) and foreign direct investments spreading over various industries, (2) overall monetary conditions remained supportive, with a broad-based corporate loan expansion, and (3) long-term public investments in infrastructure were more concrete and would crowd in private investment going forward.

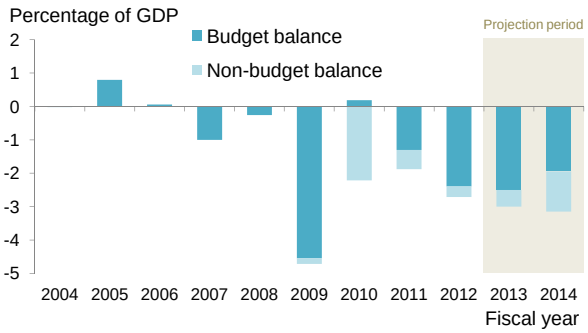
Some key risks to private investment needed monitoring. These included (1) the labor shortage problem which was the main constraint on capacity expansion and could cause some businesses to move production to neighbor countries. Tight labor market conditions persisted and were reflected through the higher vacancy to unemployment ratio and business surveys which indicated that labor recruitment was a major drawback for businesses. (2) Profit margin of businesses which were affected by the minimum wage hike and the appreciation of the baht in previous periods. This could put constraints on short-term investment decisions, especially for SMEs that did not hedge against foreign exchange risks and industries with low natural hedge. And (3) fragile recovery of major industrialized economies, which could affect investment decisions of export-oriented industries.

Continuous fiscal stimulus

Fiscal stimulus was likely to increase in the following fiscal years.

Fiscal policy would continue to provide economic stimulus going forward as reflected by the increasing trend of fiscal deficit which was relatively high in proportion to GDP. The budget deficit for fiscal years 2013 and 2014 were expected to stand at 300 and 250 billion baht, respectively. Meanwhile, non-budgetary expenditures were likely to increase over the projection period **(Chart 1.11)**.

Chart 1.11 Assumption on public sector expenditure



Note: Non-budgetary spending includes (1) the Mega Investment Project (Strong Thailand), (2) the water management project, (3) the long-term infrastructure investment project, and (4) the Development Policy Loan (DPL)

Source: Public Debt Management Office and Bureau of the Budget and calculation by Bank of Thailand

Fiscal stimulus was expected to increase from the previous projection (Table 1.4). Budget expenditures were expected to rise slightly from the increase in funds from fiscal year 2012 in line with actual disbursement data. Meanwhile, non-budgetary expenditures were also likely to increase. The MPC also incorporated government spending under the 2 trillion infrastructure project into the baseline case, with gradual disbursements expecting to commence in the beginning of fiscal year 2014. At the same time, investments in water management declined due to the slight downward adjustment in the amount of fund and the postponement of some spending on this project to later fiscal years, to be consistent with the latest project details and data on the government's disbursement plan.

Table 1.4 Assumptions on public sector expenditure

Unit: Billion baht	Fiscal year	
	2013	2014
General government consumption	1,628.0	1,766.7
Public investment	792.0	899.5
Total	2,420.0	2,666.2

Note: Includes expenditure assumptions on the water management project and the infrastructure investment project

Source: Bureau of the Budget and calculation by Bank of Thailand

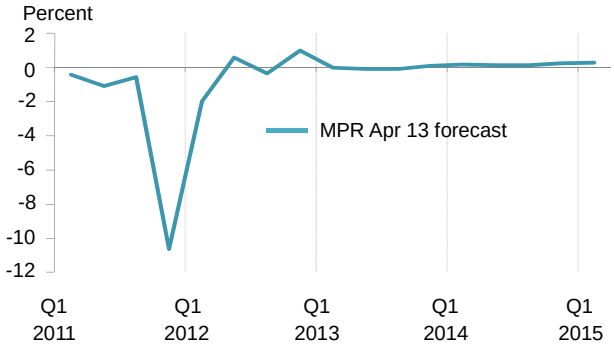
The MPC deemed that non-budgetary spending as well as contingent obligations from quasi-fiscal activities via specialized financial institutions (SFIs) warranted monitoring, as they could put constrain on future fiscal policy implementation and could affect sovereign debt prospects and fiscal position in the long run.

Inflationary pressure was likely to pick up.

Inflationary pressure should remain relatively stable during the start of the year, but was likely to pick up gradually going forward. This was consistent with the improving trend of economic conditions and the higher-than-expected world crude oil prices. Risks to inflation were slightly skewed upwards.

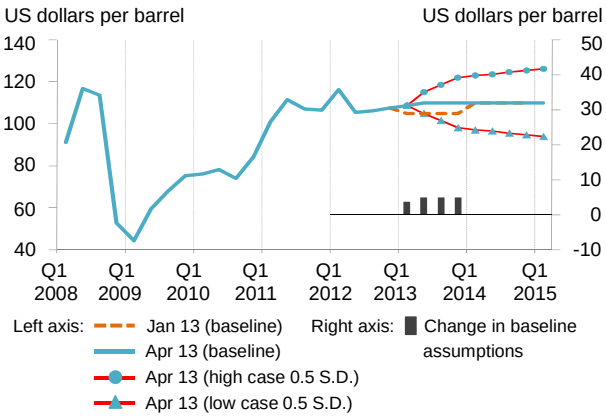
Inflationary pressure should pick up going forward, with demand pressure likely to increase steadily in tandem with the economic outlook. The MPC assessed that household spending would still gain support from favorable income prospects and accommodative fiscal and monetary conditions. Meanwhile, businesses were likely to invest continuously to accommodate demand for goods and to upgrade their manufacturing production process. Public spending on large-scale infrastructure investments also became more concrete. The gradual increase in demand pressure was consistent with the output gap becoming slightly positive in future periods **(Chart 1.12)**, though the output gap would still remain close to zero over the projection period.

Chart 1.12 Output Gap



Cost pressure was likely to increase slightly from the previous projection, in line with higher global energy prices at the start of the year. The MPC adjusted upwards the baseline assumption on the Dubai oil price to stabilize at 110 US dollars per barrel in the beginning of 2013 from the previous assumption of 105 US dollars per barrel **(Chart 1.13).** This adjustment was on account of intensified geopolitical conflicts in the Middle East and North Africa, increased global demand for crude oil following US economic recovery, as well as China and Japan's economic stimulus measures. Overall, however, world oil prices were expected to remain stable throughout the projection period. The MPC assessed that both the demand-side and supply-side

Chart 1.13 Assumptions on Dubai oil price



fundamentals were balanced and thus set high and low case assumptions on the Dubai oil price to be at half a standard deviation above and below the baseline path. With regard to domestic oil prices, the MPC continued to assume the diesel excise tax exemption throughout the projection period.

Meanwhile, cost pressure in other areas was not yet a concern. World non-fuel commodity prices **(Chart 1.14)** were expected to expand at a low level throughout the projection period, similar to that previously assumed as demand and supply conditions remained unchanged. Assumptions on domestic fresh food prices were also similar to the previous assessment **(Chart 1.15)**. On the one hand, fresh meat and seafood prices were likely to rise in the short-term from a surge in shrimp prices following a disease outbreak. On the other hand, this was offset by a drop in vegetable prices stemming from favorable weather conditions which led to more-than-expected production. Rice price remained stable.

Looking ahead, the MPC judged that upside risks to inflation would outweigh downside risks slightly because (1) there was a greater chance that shortage of labor and other production factors could put further pressure on inflation. The cost pass-through could be higher going forward given lower profit margins of businesses together with favorable consumer spending prospects that would encourage upward price and (2) geopolitical conflicts in the Middle East and North Africa could create additional pressure on world oil prices. Nevertheless, downside risks to inflation also existed including weaker private consumption than projected in the baseline case should global economic recovery be slower than that previously assessed.

Chart 1.14 Assumptions on non-fuel commodity prices

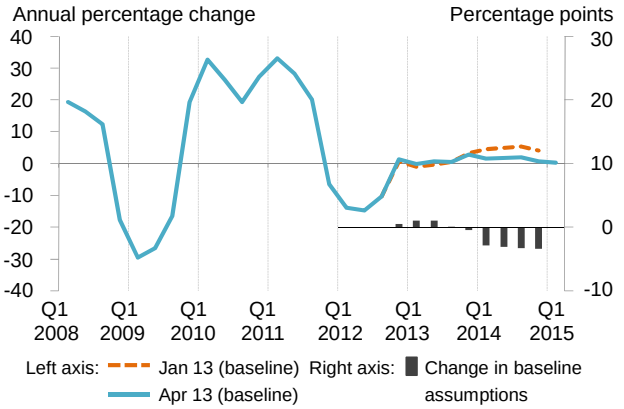
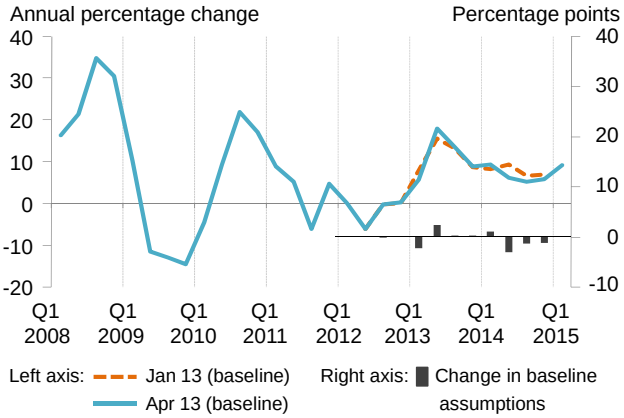


Chart 1.15 Assumptions on fresh food prices



Risks to economic stability

Overall economic stability remained sound. However, there remained risks emanating from rising household debt and more prevalent investments in risky assets.

Against the backdrop of strong Thai economic prospects, there started to be some signs of imbalance buildup in the economic system, which rendered close monitoring to prevent it from affecting the overall Thai economy. Going forward, monetary policy implementation would have to be weighed carefully between these risk factors and economic and inflation prospects.

Once source of risks to financial stability stemmed from persistently high consumer credit growth, especially in auto leasing and other personal loans. This led to an increase in household debt and could cause financial vulnerability in the household sector going forward. Although households' debt servicing ability remained intact, as reflected by a low delinquency ratio, there began to be signs of increasing stress in delinquencies for some loan types. The MPC thus deemed it necessary to continue monitoring households' debt creation and debt servicing ability.

Supportive economic and monetary conditions encouraged investments in risky assets, as reflected by high expansions in the stock and real estate markets during the recent periods. The stock market started to display signs of speculation, partly reflected by a surge in trading activity by domestic retail investors which were found to mostly speculate on small stocks. Meanwhile, both demand and supply in the real estate market was expected to expand continuously, particularly in large provinces where land and house prices were likely to rise steadily in line with increasing purchasing power

and urbanization. Some areas started to see more housing unit reservations for the purpose of investment and speculation than in the past. Real estate activities in Bangkok and its vicinity also rose. The MPC assessed that real estate risk would warrant monitoring as it could affect economic activities broadly and would take time to recover.

Alternative scenarios

The MPC appraised that there were various scenarios where Thai economic growth could deviate from the baseline case. The MPC thus took these scenarios into consideration to reflect key risk factors on the Thai economy and inflationary pressure going forward.

Scenario 1: The global economy recovered slower than assessed. Although uncertainties in the global economic system clearly declined in the recent period, global money markets were still susceptible to short-term situations. There was also a possibility that economic problems could spread out and protract. The MPC appraised that the global economy could recover slower than the baseline case should (1) austerity measures in the euro area affect demand more than expected, and problems in Cyprus spillover to Greece and other countries, (2) fiscal stimulus in the US decline and affect economic activities by more than expected and (3) Japan's stimulus measures fail to achieve the target. Under these conditions, the MPC assessed that the Thai economy would expand lower than expected in the baseline case due to the following reasons. (1) The export sector would be directly affected by weakening global demand and its recover would be pushed back to be later than the baseline projection at end-2013. (2) The corporate sector could delay investment decisions. And (3) given excess liquidity in the global market, economic problems outside the region could encourage capital flows into the region and exert upward pressure on regional

currencies, including the baht. Although the stronger currency could encourage imports and partly reduce inflationary pressure from imported inflation, it would also have an adverse effect on export value in baht terms.

Scenario 2: Fiscal stimulus could be higher than expected if the government was able to disburse the 2 trillion baht infrastructure investment project as planned. In the baseline case, the MPC appraised that disbursements, especially during the beginning of the project, would not be as high as planned by the government. If the government was able to disburse in accordance to its target plan, then economic growth in 2014 could be higher than projected in the baseline case, due to the direct increase in public investment and its positive effects on private investment and confidence.

Under the first scenario, the MPC appraised that a continuation of an accommodative monetary policy stance would be appropriate. The MPC would take a wait and see approach and could postpone interest rate normalization for some time. For the second scenario, further fiscal stimulus could cause domestic production factors to tighten and exert additional pressure on inflation. This could instigate earlier policy interest rate hikes than assumed in the baseline case going forward.

1.3 Monetary policy decision

Monetary policy stance remained accommodative.

The MPC deemed the current accommodative policy rate appropriate given fragile global economic conditions and domestic financial stability risks. The MPC thus voted to maintain the policy rate in its last two meetings.

In its meeting on February 20, 2013, the MPC assessed that the global economy became more stable and exhibited stronger signs of improvement compared to the previous meeting. However, risks from lingering resolution of the euro area's sovereign debt problems and uncertainty regarding US fiscal consolidation remained. In 2012 Q4, the Thai economy expanded more than expected and would likely grow at a higher rate than previously assessed, with domestic demand remaining the main engine of growth. In addition, exports were expected to recover gradually, while inflationary pressure increased somewhat from the previous meeting on the back of higher oil prices. The MPC deemed that the accommodative monetary policy stance in the previous periods contributed to sustained growth of the Thai economy, while inflation remained within the target range. However, uncertainties surrounding the global economic outlook and risks to domestic financial stability from rising asset prices remained. The MPC thus voted 6 to 1 to hold the policy rate at 2.75 percent per annum, with 1 vote in favor of a 0.25 percent cut to mitigate risks from capital inflows and still fragile economic momentum. Going forward, the MPC would continue to closely monitor risks to financial stability and the capital flow situation, and stand ready to take appropriate policy actions as warranted.

In its subsequent meeting on April 3, 2013, the MPC assessed that the global economy continued to recover gradually. However, tail risks had edged up somewhat since the last meeting, as financial crisis in the euro area could exacerbate economic contraction. Although the US economic outlook had improved, fiscal sequestration weighed on growth. Meanwhile, the Japanese economy displayed signs of improvement thanks to monetary and fiscal policy stimulus. The Thai economy was projected to moderate toward the normal trend in 2013 Q1, after accelerating in the previous quarter. In the periods ahead, domestic demand should continue to be a key engine for growth. Fiscal stimulus should also gradually pick up pace in the second half of this year from public investment in flood management and large-scale infrastructure projects, lending further impetus to private investment. Exports should expand slowly, in line with global economic conditions. Meanwhile, current inflation remained within the target, but potential upward pressure from supply constraints and higher labor costs would warrant monitoring. The MPC judged that, given the fragile state of the global economy, continuation of an accommodative monetary policy stance remained appropriate. However, risks to financial stability, including volatile exchange rate and capital flows, were a concern. The MPC thus voted 5 to 1 to maintain the policy interest rate at 2.75 percent per annum, with one member voting for a cut in the policy rate by 0.25 percent per annum (one MPC member was unable to attend the meeting due to obligations abroad). The MPC would continue to closely monitor domestic economic developments as well as risks to financial stability, and stand ready to take appropriate policy action.

1.4 Appendix:

Table 1.5 Forecast assumptions			
	2012	2013	2014
Dubai oil price (U.S. dollars per barrel)	108.9	110.0	110.0
Non-fuel commodity prices (%YoY)	-9.8	1.0	1.6
Fresh food prices (%YoY)	-1.5	11.4	6.6
Minimum wage in the Bangkok Metropolitan Region (baht per day)	279	300	300
Government consumption (%YoY)[1/]	10.4	8.0	6.6
Public investment (%YoY)[1/]	12.2	27.0	11.3
Fed Funds rate (% at year-end)	0.13	0.13	0.13
Trading partners' economic growth (%YoY)[2/]	3.3	3.5	4.0
Regional currencies vis-à-vis the US dollar (Index)[3/]	108.4	106.1	104.1

Note: [1/] Including spending on water management plans and infrastructure investment projects
[2/] Weighted by each country's share in Thailand's total exports
[3/] Appreciation against the US dollar indicated by a decrease

Table 1.6 GDP growth forecasts by research houses		
	2013	2014
HSBC	6.8	5.2
UBS	6.0	4.5
Credit Suisse	5.9	5.0
FPO[2/]	4.8-5.8	n.a.
BOT	5.1	5.0
Barclays	5.0	5.5
DBS	5.0	5.0
NESDB[1/]	4.5-5.5	n.a.
Siam Commercial Bank	4.9	5.3
Kasikorn Research	4.8	5.0
Tisco Securities	4.7	5.0
Phatra Securities	4.5	5.0
Standard Chartered	4.0	5.5

Note: Compiled and published by Reuters on April 3, 2013, except:
[1/] Published on February 18, 2013, with the release of GDP data for 2012 Q4
[2/] Published on March 29, 2013
Presented in descending order of 2013's forecast

Table 1.7 Headline inflation forecasts by research houses		
	2013	2014
Credit Suisse	3.7	3.9
Barclays	3.6	3.8
DBS	3.6	3.7
UBS	3.5	3.1
Tisco Securities	3.3	3.2
Kasikorn Research	3.3	3.0
HSBC	3.2	3.6
Phatra Securities	3.2	3.0
Standard Chartered	3.1	3.6
Siam Commercial Bank	3.0	3.0
NESDB[1/]	2.5-3.5	n.a.
FPO [2/]	2.5-3.5	n.a.
BOT	2.7	2.7

Note: Compiled and published by Reuters on April 3, 2013, except:
[1/] Published on February 18, 2013, with the release of GDP data for 2012 Q4
[2/] Published on March 29, 2013
Presented in descending order of 2013's forecast

Table 1.8 Probability distribution of GDP growth forecast									
Percent	2013				2014				2015
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1
> 12	0	0	0	0	1	1	1	1	2
10-12	3	0	1	1	3	3	4	4	5
8-10	30	3	7	4	10	11	12	12	12
6-8	51	17	23	14	21	21	22	20	20
4-6	15	37	34	27	28	27	26	25	24
2-4	1	32	24	30	23	21	20	20	20
0-2	0	10	8	18	11	11	11	11	12
(-2)-0	0	1	1	6	3	4	4	4	5
< (-2)	0	0	0	1	1	1	1	1	2

Table 1.9 Probability distribution of headline inflation forecast								
Percent	2013			2014				2015
	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1
> 7	0	0	0	1	1	2	2	3
6-7	0	0	1	2	3	4	4	5
5-6	2	3	5	6	8	9	9	10
4-5	13	11	11	12	14	15	15	15
3-4	36	24	19	19	20	20	19	19
2-3	37	31	24	22	21	20	20	19
1-2	12	22	21	19	17	16	15	15
0-1	1	8	12	11	10	9	9	9
(-1)-0	0	2	5	5	4	4	4	4
< (-1)	0	0	1	2	2	2	2	2

Table 1.10 Probability distribution of core inflation forecast								
Percent	2013			2014				2015
	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1
> 4.0	0	0	0	1	1	1	1	1
3.5-4.0	0	0	1	2	2	2	2	2
3.0-3.5	0	1	4	6	6	5	6	6
2.5-3.0	1	7	11	13	13	11	11	11
2.0-2.5	11	21	21	21	20	18	18	17
1.5-2.0	37	33	26	24	23	22	21	20
1.0-1.5	40	26	22	19	19	20	19	19
0.5-1.0	11	9	11	10	11	13	13	13
0.0-0.5	1	1	3	3	4	6	6	7
< 0.0	0	0	1	1	1	2	3	3

What are the determinants of capital inflows to Thailand?

The issue of capital inflows, particularly inflows into the bond and stock markets, is currently receiving much attention given the recent sharp increases in their size and speed. In this regard, many are concerned about their consequences including the volatility of the flows themselves and the exchange rate, which could negatively affect the country's economic and financial stability.

In the past, many studies of determinants of capital flows have relied on two types of economic variables as follows; (1) pull or domestic factors such as gross domestic product, inflation, and interest rate differential; and (2) push or external factors such as global economic growth and liquidity in major industrialized financial markets. However, changes in the global environment since the latest global economic crisis and the euro area's sovereign debt problems have shifted the view and behavior of investors. Specifically, in a world of volatile capital movements and exchange rates, investors are highly sensitive to global uncertainties and tend to take more short-term speculative positions. Therefore, a study of capital inflow determinants that only takes into account data on investment returns and economic fundamentals may prove inadequate. Factors that reflect expectation and sentiment on risks should also be considered.

Determinants of capital inflows into the Thai stock market[1/]

This article studies determinants of capital inflows into the Thai bond and stock markets using flow data on net purchase of the Thai baht for equity investment by financial institutions and other foreign investors. This dataset is a more accurate measure of capital inflows for the purpose of this study than balance of payments data which was often used in prior studies because it is a record of actual payments on a cash basis. Therefore, this dataset can reflect new flows to the country as well as exchange rate pressure more accurately than balance of payments data which records ownership changes between residents and nonresidents without any regard to actual cash settlements.

An important assumption in this study is that investors seek high returns but dislike risks. Thus, they consider factors such as returns on investment in stocks and bonds, interest rates, and exchange rate expectation. A regression analysis is performed using data from July 2007 to September 2012. The regression uses eight independent variables that can be classified into two groups as follows; (1) macroeconomic variables, including the Purchasing Managers Index (PMI), the Business Sentiment Index (BSI), policy interest rate differential between Thailand and the US, Thai unemployment rate, and Thai inflation; and (2) variables reflecting investors' sentiment, namely VIX index, expected Thai baht appreciation and Thai baht implied volatility.[2/]

The study found that changes in investors' sentiment on risks, expected volatility of the Thai baht and Thai inflation had negative effects on foreign inflows into the bond and stock markets. On the other hand, expected growth of the Thai economy, expected baht appreciation and changes in policy interest rate differential between Thailand and the US had positive effects on inflows **(Table 1 Model A)**. The results are consistent with the intuition that foreign investors seek high returns from (1) interest rates; (2) corporate profitability that is expected to be healthy in

---

[1/] Results from FAQ Issue 78 "Do interest rate differentials affect capital inflows?" by Supachoke Thawornkaiwong and Chidchanok Annonchan published on the Bank of Thailand's website on March 4, 2013 (in Thai).

[2/] Purchasing Managers Index (PMI) is a survey of final production of major businesses which reflects global economic growth. Business Sentiment Index (BSI) is a survey of businesses which reflects businesses' view on Thailand's expected economic growth VIX reflects sentiment on risks of global investors, which is available from the Chicago Board Options Exchange Implied volatility of the Thai baht reflects expected volatility of the Thai baht based on information from foreign options markets

line with a strong economy; and (3) speculation on the Thai baht, while risks and volatility are shunned by investors.

Table 1 Coefficients of determinants of capital inflows into the Thai bond and stock markets

Variables	Model A	Model B
Intercept	-0.0731	-0.3334
Changes in VIX	-0.3714*	-0.3192*
Changes in global PMI	0.2100	0.0237
BSI (1-month lag)	0.0054*	0.0097*
Unemployment (1-month lag)	-0.0115	-0.0066
Inflation (1-month lag)	-0.1296*	-0.0925*
Changes in interest rate differential	0.1706*	
Interest rate differential		0.0009
Thai baht appreciation	0.6208*	0.6593*
Thai baht implied volatility	-0.0192*	-0.0114*
R^2	0.61	0.56

* Statistically significant

Table 2 Relative importance of determinants of capital inflows into the Thai bond and stock markets

Statistically important variables	Ability to explain foreign capital inflows into the Thai stock market (percent)
Changes in VIX (reflects investors' sentiment)	18
Outlook for Thai baht appreciation	15
BSI (reflects expected economic growth)	11
Expected volatility of the Thai baht	10
Changes in interest rate differential	3
Inflation	3

However, it is interesting to note that the results were not consistent with the theory of interest rate parity, which explains investors' behavior based on their beliefs that assets of two countries are perfect substitutes such that they will always invest in assets with higher return. In other words, this study does not find that policy interest rate differential between Thailand and the US was the key determinant of inflows into the stock market. However, changes in investors' sentiment on risks and changes in policy interest rate differentials do matter. Further regression analysis found that the level of Thai and US interest rates have very small coefficients and are not statistically significant **(Table 1 Model B)**.

This study also analyzed the relative contribution of the explanatory variables that were statistically significant. A statistical method of computing relative importance was applied to the results and we found that the sentiment variables, namely changes in investors' sentiment on risks and expected Thai baht appreciation, were more statistically important in determining movements of foreign capital into the Thai bond and stock markets than macroeconomic variables **(Table 2)**.

Conclusion and policy implication

Capital inflows into the Thai bond and stock markets depend on many variables, both economic and sentiment variables. With regard to implementation of the interest rate policy, the study suggests that reducing Thailand's interest rates in an effort to deter foreign inflows would render only a small and short-term effect. This is because investors are responsive only to new information. The findings implied that if the interest rate differential between Thailand and the US was left unchanged in the next periods and all other economic variables remained the same, foreign investors would not change their investment position. This means that, holding everything constant, the effect of a policy rate cut on capital inflows would last for only one period. Therefore, policy makers need to carefully weigh the benefits of a policy interest rate cut to stem inflows in the short-term against the costs of negative effects from a possible inconsistency between the level of policy rate and the economic conditions.

Recent Economic Developments

2. Recent Economic Developments

The world economy recovered gradually on the back of an improvement in private consumption and investment in the US, as well as Japan's economic stimulus measures and strong domestic demand growth in China and Asian economies. Nevertheless, euro area economies remained weak on the back of political uncertainty in Italy and the financial crisis in Cyprus which likely affected financial markets and economic recovery in the short run.

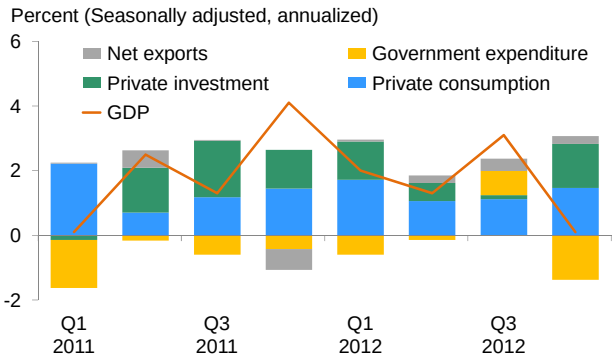
The Thai economy in 2013 Q1 was expected to moderate towards the normal trend after a strong acceleration in the previous periods. Economic fundamentals and confidence remained intact, contributing to a continuation in private spending and investment growth. At the same time, fiscal stimulus continued to lend support to domestic demand, while the export sector improved slowly in line with gradual global economic recovery. Current inflation remained in the target range. However, inflationary pressure from supply constraints and higher labor costs warranted monitoring.

2.1 The global economy

The US economy was expected to improve on the back of strong private consumption and investment growth, thanks to improving employment conditions and clearer signs of recovery in the housing sector. Fiscal problems, however, remained a key risk to the economy.

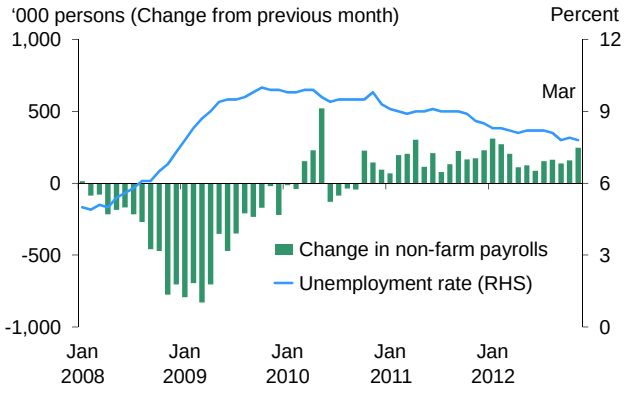
In 2012 Q4, the US economy overall remained unchanged from the previous quarter. **(Chart 2.1)** owing to two offsetting factors. On the one hand, public spending contracted, especially on national defense and inventory accumulation. But the contraction in these categories were expected to be temporary. On the other hand, growth in private consumption and investment accelerated, with much clearer signs of recovery in housing sector investments and some incipient signs of a pickup in corporate investments.

Chart 2.1 Contribution to US economic growth
(Change from the previous quarter)



Source: Bureau of Economic Analysis

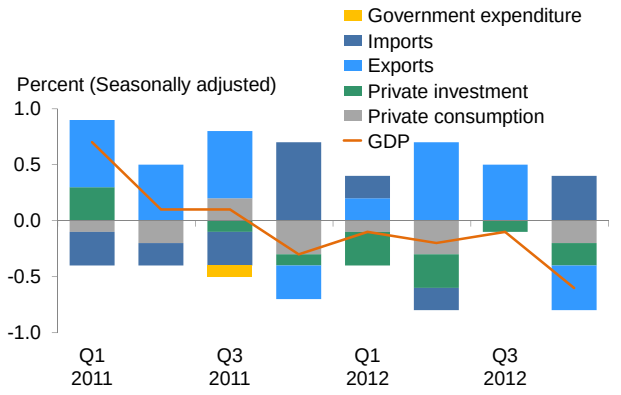
Chart 2.2 US labor market conditions



Source: Bureau of Labor Statistics

Chart 2.3 Contribution to the euro area's economic growth

(Change from the previous quarter)



Source: Eurostat

In 2013 Q1, the US economy was likely to recover gradually spurred by improvement in private consumption as reflected by high automobile sales and retail sales growth. Higher employment contributed to this more positive outlook **(Chart 2.2)**, despite the increase in the personal income tax rates and payroll tax since January 2013. Meanwhile, private investment was likely to increase thanks to continual recovery in the real estate sector and improving business confidence.

The MPC assessed that the US economic outlook would likely improve in the next periods, led mainly by the increasing trend of private consumption and investment. However, fiscal sequestration remained the main constraint on economic growth. Going forward, uncertainties regarding the debt ceiling remained, and there was a possibility that the US government would have to further reduce the budget deficit in future periods. Prolonged and unclear resolution of the fiscal problems could affect business and household confidence and hence could delay private investment and spending.

Euro area economies remained weak and a slow recovery was expected in 2013. Political uncertainty in Italy would affect economic recovery but only in the short run, while risks to the economy increased from Cyprus' approach to resolve its financial crisis.

In 2012 Q4, the euro area economy contracted by 0.6 percent from the previous quarter **(Chart 2.3)** owing to a contraction in exports from the global economic slowdown. In addition, private consumption and investment weakened due to investors' and consumers' concerns over the impact from intensified sovereign debt problems in Greece and Spain.

In 2013 Q1, the euro area economy was expected to contract, albeit at a lower rate, from the previous quarter on the back of a lower contraction in

manufacturing production from the previous quarter, while exports to countries both outside and within the euro area improved. Moreover, in the first quarter, producer and consumer confidence in the economy increased from the previous quarter **(Chart 2.4)**. Meanwhile, political issues in Italy could have short-term impacts on euro area economies, but would not develop into a systemic risk. This issue, however, would be closely monitored.

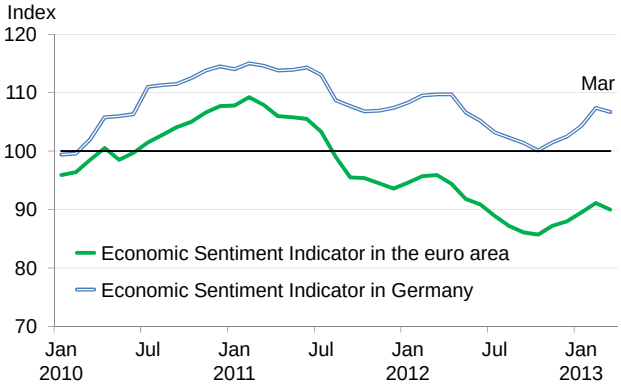
The MPC appraised that euro area economies in the first half of 2013 were likely to weaken more than expected, before recovering slowly in the latter half of the year. Risks to the euro area increased on the back of Cyprus' approach to resolve its financial crisis which raised concerns by other member states that the euro area authorities could also enforce this type of measure should problems in the financial sector arise in their economies.

The overall Japanese economy remained weak, but exhibited signs of stabilization. Improvement in economic performance was expected in future periods supported by government stimulus measures and the recovery of trading partners' economies.

In 2012 Q4, the Japanese economy remained more or less unchanged from the previous quarter, after a contraction of 0.9 percent in the third quarter **(Chart 2.5)**. Key positive contribution stemmed from private consumption, particularly in non-durable goods, and public spending following the additional stimulus measures from the remaining fiscal year 2011.

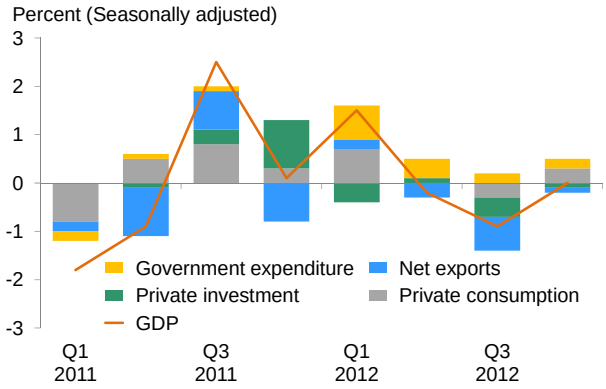
In 2013 Q1, Japan's economy remained weak, but started to show signs of stabilization as reflected by an increase in export volume in January and February in line with trading partners' economic recovery and continued improvements in producer

Chart 2.4 Economic Sentiment Indicator in the euro area and Germany



Source: Bloomberg

Chart 2.5 Contribution to Japan's economic growth
(Change from the previous quarter)



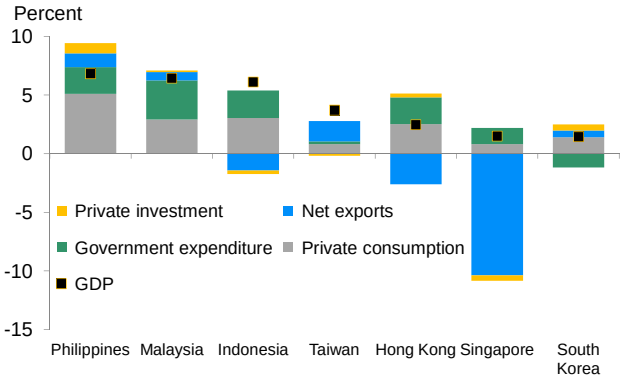
Source: Cabinet Office of Japan

and consumer confidence. This was partly due to fiscal stimulus measures and the accommodative monetary policy stance in Japan. To help pull the Japanese economy out of deflation, on January 22, 2013, the Bank of Japan adjusted the inflation target (excluding raw food) to stand at 2 percent from the previous target of 0-2 percent, and announced an open-ended asset purchase program at a pace of 13 trillion yen per month, beginning January 2014.

The MPC appraised that the Japanese economy would recover slowly on the back of continual support from government policies through additional fiscal spending and a more accommodative monetary policy stance. These would help improve corporate performance and private confidence, which would lend support to the real economy in the periods ahead.

Chinese and most Asian economies were likely to continue to expand well given strong domestic demand growth, while exports would recover gradually.

Chart 2.6 Contribution to Asia's economic growth in 2012 Q4 (Change from the same period in the previous year)



Source: CEIC

In 2012 Q4, most Asian economies improved from the previous quarter **(Chart 2.6)** following strong growth in domestic consumption and investment as well as healthier exports.

Regional economic activities in 2013 Q1 were likely to expand continuously, with consumption and investment remaining the main engine of growth. In addition, exports improved gradually, especially exports of electronics which recovered in tandem with the global electronics cycle.

The MPC assessed that regional economies would continue to expand well, mainly supported by strong domestic demand. Furthermore, exports were expected to improve gradually in line with global economic recovery and robust economic expansion in China which potentially had large influence on the

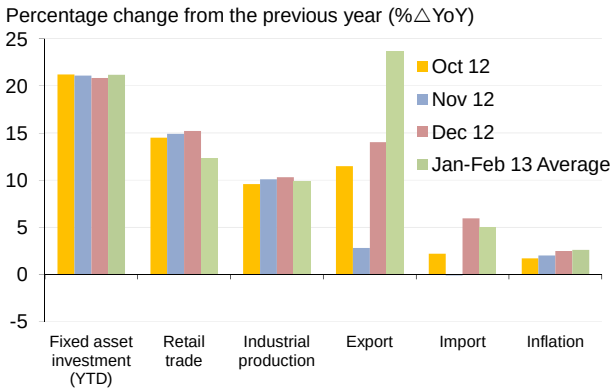
other Asian economies due to its large size and an its role as a key trading partner.

In 2012 Q4, the Chinese economy grew year-on-year by 7.9 percent, driven mainly by domestic demand, both from robust private consumption and investment, as well as continued recovery in exports after experiencing a slowdown in the first half of 2012 **(Chart 2.7)**.

In 2013 Q1, China's economic activities, including consumption, investment, production and exports, continued to expand strongly. Nevertheless, signs of imbalances in the real estate sector started to emerge as reflected in accelerating housing prices that had been rising continuously for several months during the beginning of 2013. As a result, on March 1, 2013, the government announced further tightening measures in an attempt to stabilize prices in the real estate market.

Going forward, the MPC judged that the Chinese economy would expand steadily, driven by the increasing role of domestic demand, as well as continuous infrastructure investments by the public sector, while support from the export sector remained strong.

Chart 2.7 China's economic indicators

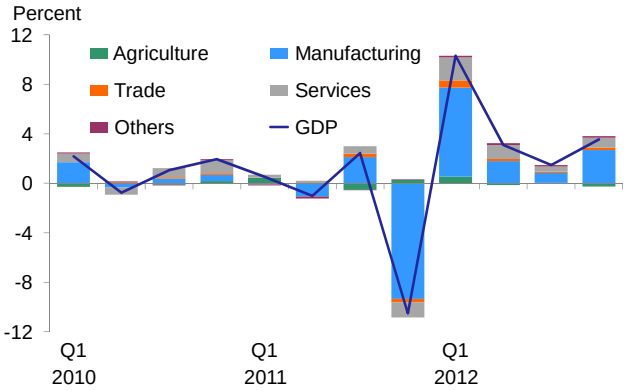


Source: CEIC

2.2 The domestic economy

In 2012 Q4, the Thai economy expanded from the previous quarter, driven mainly by continued strong growth in domestic demand, both consumption and investment of the private and public sectors. This was partly due to fiscal stimulus measures, in particular the first-car tax rebate scheme. Moreover, gradual recovery of the global economy drove exports and export-oriented manufacturing production to increase from the previous quarter.

Chart 2.8 Contribution to GDP growth
(QoQ, seasonally adjusted)



Source: National Economic and Social Development Board

Economic conditions in 2012 Q4.

In the fourth quarter, the seasonally adjusted economic growth rate for Thailand stood at 3.6 percent from the previous quarter **(Chart 2.8)** due mainly to expansions in the manufacturing and services sectors. Compared with the previous quarter, seasonally adjusted **manufacturing production** rose by 6.9 percent, consistent with recovery in export-oriented production. The hard disk drive industry recovered in line with external demand, while the integrated circuit industry began to increase production after the flood problems subsided. Meanwhile, production for domestic consumption remained strong, in particular automobile production which grew exceptionally well in response to a surge in car orders incentivized by the first-car tax rebate scheme. Furthermore, production of the petroleum industry rose once an oil refinery resumed production after it was shut down for repairs due to a fire in the previous quarter.

The **services sector** expanded continuously at 2.3 percent from the previous quarter, on the back of strong growth in the trade sector following domestic demand conditions. This in turn, led to expansion in the logistics sector. The construction sector grew both in the private and public sector. Construction of the private sector was mainly for the purpose of production capacity enhancement and investments in real estate projects in Bangkok and other large provinces, while construction of the public sector was carried out by both the central and local governments. The tourism sector grew satisfactorily in tandem with a surge in the number of foreign tourists. As a consequence, other tourism-related services such as hotel, restaurant and transportation, also expanded well.

Seasonally adjusted **agricultural outputs** contracted from the previous quarter by 3.1 percent, after accelerating robustly in the third quarter which experienced unusually high expansion in second crop

rice production due to an increase in planting area following price incentives from the rice pledging scheme. Shrimp output declined as a result of a disease outbreak in some areas in the eastern part of Thailand. Other agricultural outputs such as rubber and palm oil expanded due to increased planting area, while cassava output increased on the back of favorable weather conditions.

Continual economic expansion led to a low unemployment rate of 0.5 percent in 2012 Q4. Employment in the agricultural sector grew in tandem with the planting season, while employment in non-agricultural sectors expanded mostly in the manufacturing, construction and trade sectors.

Demand in 2012 Q4 expanded on the back of continued private consumption growth. Meanwhile, exports showed clearer signs of recovery, especially towards the end of the quarter. This was consistent with gradual improvements in export-oriented manufacturing production.

Table 2.1 GDP growth rate

Change from the previous quarter (seasonally adjusted, percent)	2012				
	Year	Q1	Q2	Q3	Q4
GDP	6.4	10.3	3.1	1.5	3.6
Domestic demand[1/]	8.4	7.3	2.6	2.9	2.1
Private consumption	6.6	6.5	1.0	2.1	2.6
Private investment	14.6	12.8	4.4	4.3	-0.3
Government consumption	7.4	3.8	6.5	3.0	-0.8
Public investment	8.9	2.9	6.1	6.3	13.0
Exports of goods and services	2.9	10.3	3.6	1.1	3.1
Imports of goods and services	6.2	7.6	4.7	-2.9	5.2

Note: [1/] Domestic demand excluding changes in stocks
Source: National Economic and Social Development Board and calculations by Bank of Thailand

Private consumption expanded by 2.6 percent from the previous quarter **(Table 2.1)** following strong demand growth for durable and non-durable goods, particularly automobiles. Supporting factors

included government stimulus packages, satisfactory employment conditions and higher household income following the minimum wage hike, as well as favorable monetary conditions conducive to borrowing. Meanwhile, private investment declined 0.3 percent from the previous quarter after having accelerated in the previous quarters to replace and repair flood-related damages. Nevertheless, investments in machinery and equipment, as well as in construction remained high.

Gradual recovery of the global economy led **exports of goods and services** to expand by 3.1 percent from the previous quarter, attributable to an increase in manufacturing exports, namely automobiles, metal products, electrical appliances and electronics. Exports of agricultural products also increased in line with export volume of rice and rubber. Meanwhile, exports of services continued to expand from the previous quarter following a surge in foreign tourists, particularly from Russia, China and Malaysia, which was partly due to the reopening of Don Mueang Airport in October 2012. Furthermore, freight receipts and other transport receipts rose in tandem with higher international trade volume.

Robust domestic demand and gradual improvements in exports led **imports of goods and services** to expand by 5.2 percent from the previous quarter, thanks to a broad-based expansion in imports. The increase in imports was particularly observed in capital goods including machinery and equipment, and aircrafts, as well as imports of raw materials and intermediate goods, such as parts of both electronics and electrical appliances. This was consistent with gradual improvements in export-oriented production. Imports of services also increased from the previous quarter from freight payment which rose in line with international trade volume, and passenger transport payments following an increase in the number of Thais travelling abroad.

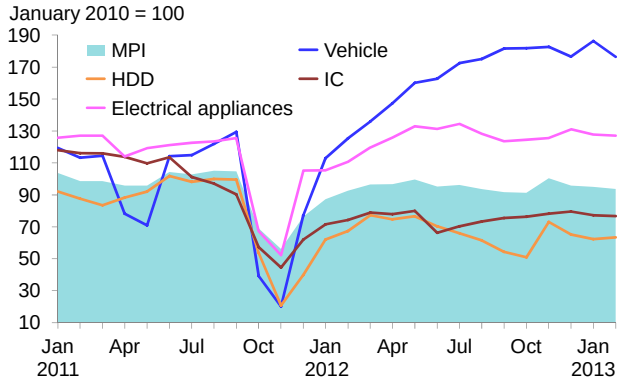
The **public sector** continued its role in stimulating the economy, as reflected by an expansion of public spending from the previous quarter, in particular investments in durable assets by the Royal Irrigation Department, the Department of Highways and the Department of Rural Roads.

The economic outlook for 2013 Q1[1/]

In 2013 Q1, the Thai economy was likely to expand slightly from the previous quarter. Private consumption and investment were expected to moderate towards the normal trends after having accelerated strongly in the previous quarters. Meanwhile, economic fundamentals remained healthy. Fiscal stimulus would continue, while exports should improve in line with gradual global economic recovery. The manufacturing sector was expected to remain unchanged from the preceding quarter due to shortage of raw materials in the food industry and capacity constraints in the automobile industry as it was already operating at full capacity since the end of last year. Nevertheless, the services sector was expected to grow continuously consistent with expansion in the trade, transportation, construction and tourism sectors.

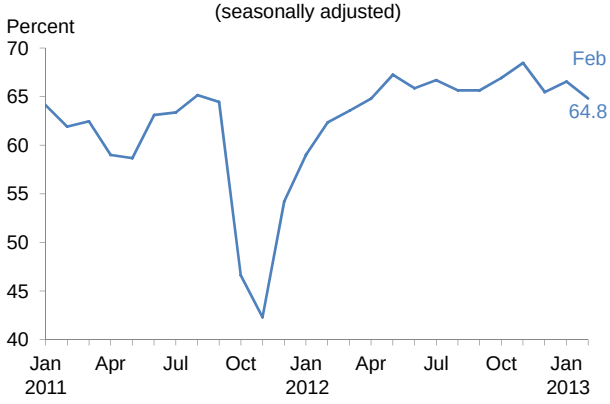
[1/] Economic indicators used in assessing the outlook for 2013 Q1 were produced by the Bank of Thailand, except data on the Manufacturing Production Index and capacity utilization rate, which were provided by the Office of Industrial Economics. The number of tourists and the occupancy rate were, in part, compiled by the Tourism Authority of Thailand. Data on the labor market were obtained from the National Statistical Office. The government expenditure data were provided by the Comptroller General's Department, which compiled by the Fiscal Policy Office, the Ministry of Finance. Economic outlook were obtained from the Economic/Business Information Exchange Program between the Bank of Thailand and business sector.

Chart 2.9 Manufacturing Production Index (MPI)
classified by sectors (seasonally adjusted)



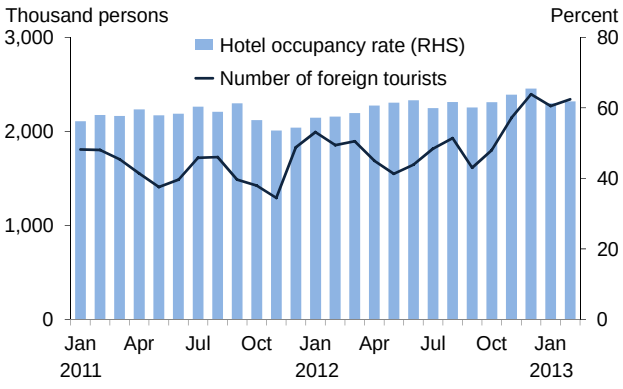
Source: Office of Industrial Economics, Ministry of Industry and seasonally adjusted by Bank of Thailand

Chart 2.10 Capacity utilization rate
(seasonally adjusted)



Source: Office of Industrial Economics, Ministry of Industry and seasonally adjusted by Bank of Thailand

Chart 2.11 Number of foreign tourists and hotel occupancy rate (seasonally adjusted)



Source: Department of Tourism and Bank of Thailand

Manufacturing production in 2013 Q1 was expected to remain stable compared to the previous quarter, constrained by from raw material shortages. The frozen seafood industry experienced shortage in shrimps due to disease outbreak problems. Sugar production was also likely to decline as a result of lower sugarcane output in this harvest season. Moreover, the automobile industry was already operating at full capacity in the previous quarter and thus could not increase production until capacity-enhancing investments were completed. Export-oriented production was expected to recover gradually in tandem with global economic conditions. The hard disk drive industry was likely to expand slightly, while overall production of integrated circuits declined following sluggish global demand, except for electronic parts used in smartphones and tablets which expanded robustly **(Chart 2.9)**. Stable manufacturing production led the capacity utilization rate in the first quarter to remain similar to that in the previous quarter **(Chart 2.10)**.

The **services sector** in the first quarter overall was expected to expand well in line with the trade, transportation, construction and tourism sectors. Wholesale and retail sales, and freight were expected to increase strongly in tandem with domestic consumption. Meanwhile, the construction sector should continue to expand robustly as a result of both public investments, such as the mass rapid transit system, and private investments such as the new automobile factory and investments in the real estate sector which were likely to expand in various provinces on account of higher purchasing power and urbanization. The tourism sector continued to grow robustly as reflected by the number of tourists and the seasonally adjusted hotel occupancy rate which rose steadily, especially tourists from China and Russia **(Chart 2.11)**.

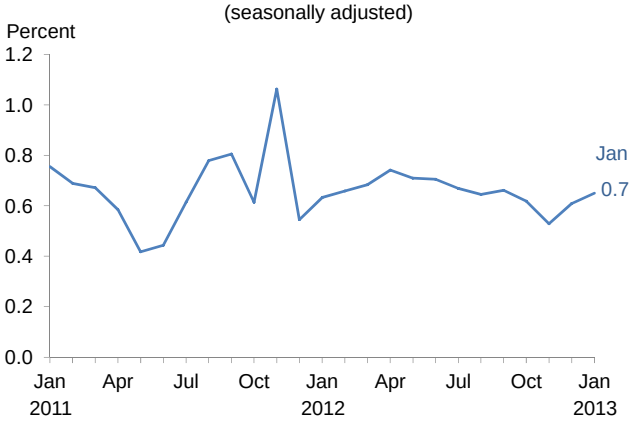
Agricultural production in the first quarter was likely to contract from the previous quarter. This was due to a decline in rice output as farmers expected a drought during the end of last year and hence quickly harvested to prepare for the new round of off-season planting. As a result, rice output in this quarter was less than usual. Meanwhile, rubber output was also likely to decline due to unfavorable weather conditions.

The **labor market** was expected to remain tight in the first quarter, consistent with the Business Sentiment Survey[2/] which found that firms experienced recruiting difficulties, with the problem expecting to intensify going forward. Consequently, the seasonally adjusted unemployment rate remained low **(Chart 2.12)**. Furthermore, the implementation of the second-round minimum wage hike to 300 baht per day, affecting nationwide, on January 1, 2013, did not yet lead to a significant increase in unemployment. This was partly because businesses had already made adjustments to mitigate impacts from higher labor costs**.** At the same time, although some businesses had laid off workers, part of these workers were able to find new jobs as most businesses still increased hiring. This was consistent with the Business Sentiment Survey which found that most businesses, in both manufacturing and non-manufacturing industries, planned to increase employment in the next 3 months.

For the demand side, in 2013 Q1, private consumption and investment were expected to expand from the previous quarter. Meanwhile, exports would recover at a gradual pace, in tandem with a still weak global economic recovery, while the public sector would continue to lend support to economic growth.

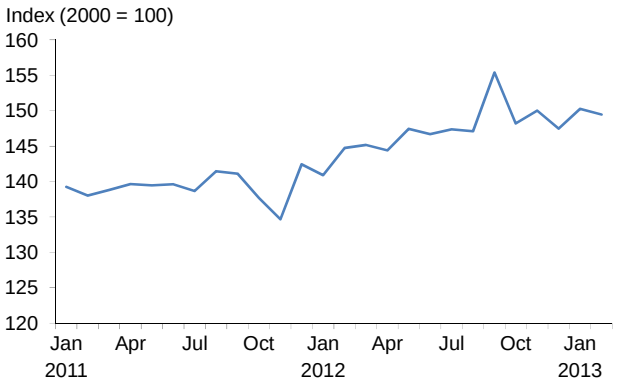
Private consumption was expected to grow at a rate close to its normal trend, as reflected by a rise in the Private Consumption Index in January and

Chart 2.12 Unemployment rate
(seasonally adjusted)



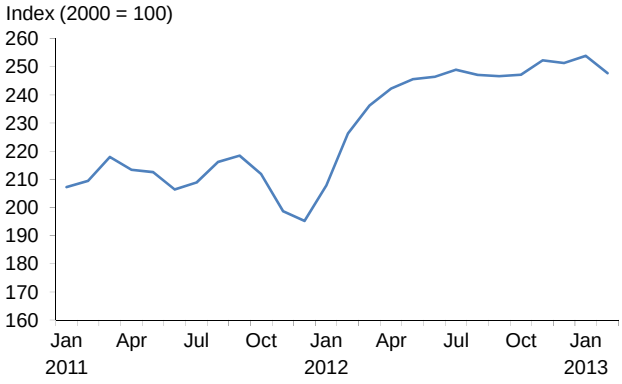
Source: National Statistical Office, and calculations by Bank of Thailand

Chart 2.13 Private Consumption Index (PCI)
(Seasonally adjusted)



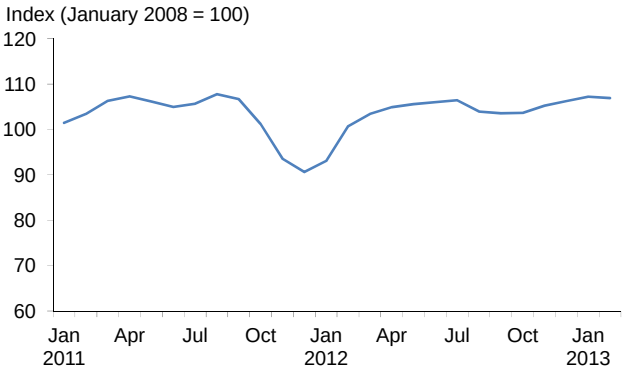
Source: Bank of Thailand

Chart 2.14 Private Investment Index (PII)
(Seasonally adjusted, 3-month moving average)



Source: Bank of Thailand

Chart 2.15 Export volume index
(seasonally adjusted, 3-month moving average)



Note: Data for January and February 2013 calculated by the Bank of Thailand
Source: Bank of Thailand

February 2013 **(Chart 2.13)**. This was mainly attributable to favorable household income and employment conditions which helped increase purchasing power and consumer confidence. Fiscal and monetary policies also played a role in supporting expansion in private consumption. This included the first-car tax rebate scheme, personal income tax rate cut, the nationwide minimum wage hike to 300 baht on January 1, 2013, the extension of the rice pledging scheme, as well as the continuation of an accommodative monetary policy stance.

Private investment was also expected to expand, but at a lower rate than the normal trend after having accelerated strongly in the previous periods. This was reflected in a slight decline in the Private Investment Index in January and February 2013 **(Chart 2.14)**. Nevertheless, investments in construction as well as machinery and equipment remained at a high level and were likely to increase further in order to replace manual labor in response to the wage increase and the labor shortage problem, as well as to accommodate higher domestic and foreign demand anticipated following a closer regional economic integration under the ASEAN Economic Community (AEC). As reflected by high levels of application and issuance of investment promotion certificates by the BOI, foreign investors continued to show their interest in investing in Thailand due to strong confidence in the Thai economic outlook as well as monetary conditions conducive to seeking funds.

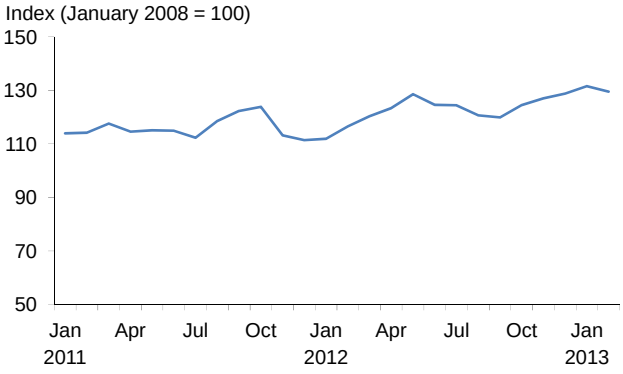
Exports of goods and services were expected to expand slightly from the previous quarter in tandem with gradual recovery of the global economy **(Chart 2.15)**. Exports of manufacturing products, especially automobiles, hard disk drives and electrical appliances were expected to decelerate from the previous quarter. Meanwhile, exports of agricultural and

fishery products were likely to decline on the back of raw material shortage, particularly shrimp for frozen seafood production, as well as deteriorating price competitiveness which should cause a slowdown in rice exports. Nevertheless, exports of services were expected to continue to expand from the previous quarter, in tandem with an increase in the number of foreign tourists from almost every region, especially China.

Imports of goods and services were likely to decelerate **(Chart 2.16)** with a slowdown in the imports of raw materials and intermediate goods, especially fuel, and a deceleration in the imports of capital goods after having accelerated in the previous periods. However, imports of consumer goods, including vehicles and parts were likely to grow strongly, in line with expansion in private spending and recovery in exports. Meanwhile, imports of services were also expected to increase from intellectual property expenditures and insurance and pension payments, as well as financial service payments following economic improvements.

Chart 2.16 Import volume index

(seasonally adjusted, 3-month moving average)



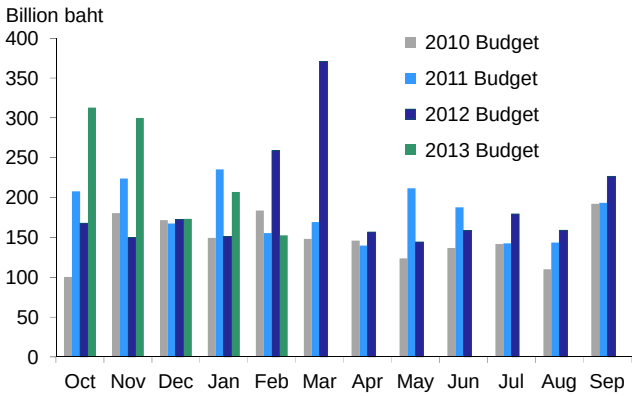
Note: Data for January and February 2013 calculated by the Bank of Thailand

Source: Bank of Thailand

Fiscal stimulus was likely to continue in the first quarter **(Chart 2.17)** as reflected by an increase in budget disbursement, from both current expenditures, especially in wages and salaries, and purchases of goods and services**,** and investment expenditures in durable assets, such as investments for flood rehabilitation and prevention, as well as investments of the Department of Highways and the Royal Irrigation Department.

In summary, the Thai economy in 2013 Q1 was poised to expand from the previous quarter on the back of increases in domestic spending, tourism and exports of goods and services in tandem with gradual improvements in manufacturing production.

Chart 2.17 Disbursement of government budget



Note: Excluding principal repayments and replenishments of the treasury reserve

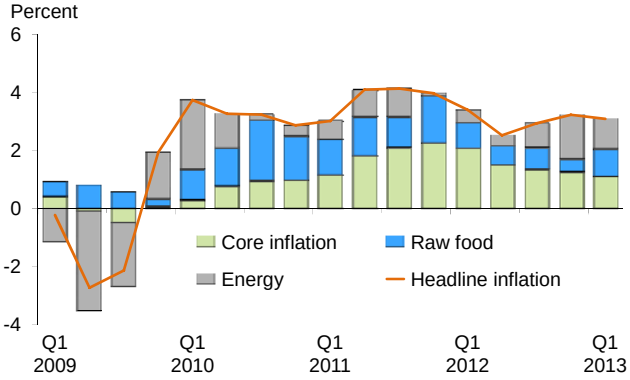
Source: Comptroller General's Department, Ministry of Finance

2.3 Costs and prices

Inflation

Headline inflation in 2013 Q1 stood at 3.09 percent, slowing down slightly from 3.23 percent in the previous quarter **(Chart 2.18)**, consistent with a slowdown in core inflation and energy prices. Energy prices in 2013 Q1 increased by 8.76 percent from the same period last year, decelerating sharply from 14.79 percent observed in the previous quarter due to a high base effect. The high base effect came from elevated domestic retail oil prices in the same period of last year following the reintroduction of the oil fund levy and higher world crude oil price from tight global supply. Nevertheless, compared to the previous quarter, domestic retail oil prices in this quarter increased in line with world crude oil prices.

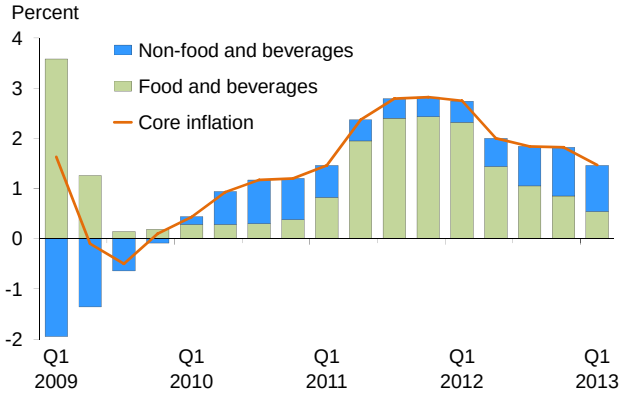
Chart 2.18 Contribution to headline inflation



Source: Trade and Economic Index Bureau, Ministry of Commerce and calculations by Bank of Thailand.

Prices of raw food rose by 6.08 percent compared to the same period last year, accelerating sharply from 2.92 percent recorded in the previous quarter, due to price increases in meat, poultry, aquatic animals and eggs. The price increases arose from (1) the relief of the excess supply problem and (2) higher demand for consumption during the Chinese New Year festival. However, prices of vegetables and fruits declined from the previous quarter on account of favorable weather conditions which helped increase output.

Chart 2.19 Contribution to core inflation



Source: Trade and Economic Index Bureau, Ministry of Commerce and calculations by Bank of Thailand.

Core inflation in 2013 Q1 stood at 1.47 percent, moderating from 1.82 percent in the previous quarter owing mainly to the slowdown in prepared food prices **(Chart 2.19)** due to the high base effect.

Table 2.2 Quarterly inflation

Unit: Percent	2011	2012				2013
		Year	H1	Q3	Q4	Q1
Percentage change from the previous year (%ΔYoY)						
- Headline Consumer Price Index	3.81	3.02	2.95	2.94	3.23	3.09
• Core Consumer Price Index	2.36	2.09	2.37	1.84	1.82	1.47
• Raw food	8.66	4.40	4.98	4.80	2.92	6.08
• Energy	5.59	7.09	3.31	7.54	14.79	8.76
Percentage change from the previous quarter (%ΔQoQ)						
- Headline Consumer Price Index	-	-	-	1.0	0.4	0.5
• Core Consumer Price Index	-	-	-	0.4	0.3	0.2
• Raw food	-	-	-	2.2	0.3	0.4
• Energy	-	-	-	2.6	1.3	2.1

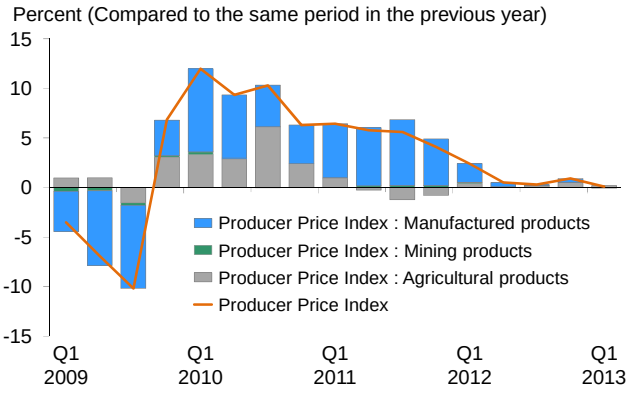
Source: Trade and Economic Index Bureau, Ministry of Commerce

There had not been signs of unusually high pass-through of higher costs from the minimum wage hike in January 2013 to prices of goods and services in 2013 Q1. This was reflected in the quarter-on-quarter core inflation which was close to the 5-year average rate **(Table 2.2)**.

Production cost conditions

Continuing from the previous quarter, overall costs of production in 2013 Q1 remained low, as reflected by an increase in the Producer Price Index of 0.1 percent, moderating from 0.9 percent recorded in the previous quarter **(Chart 2.20)**. This followed a deceleration in the price of agricultural products, mainly vegetables and fruit, after more output entered into the market. In addition, the price of petroleum products contracted due to the high base effect from high world crude oil prices on the back of tight global supply in the previous year.

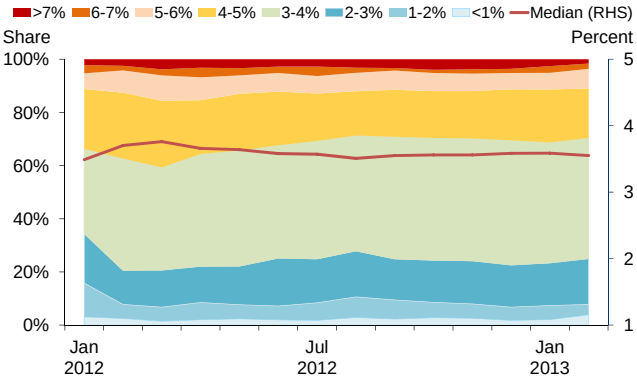
Chart 2.20 Contribution to Producer Price Index



Source: Trade and Economic Index Bureau, Ministry of Commerce, and calculations by Bank of Thailand.

Chart 2.21 12-month ahead inflation expectation

(February 2013)



Source: Bank of Thailand's Business Sentiment Survey

Inflationary pressure in the periods ahead

Inflationary pressure was likely to build up slightly in the future periods, mainly from domestic demand pressure.

(1) Demand pressure was likely to increase in line with robust economic growth, in particular private consumption and investment which were the main engine of growth, as well as gradual export recovery.

(2) Going forward, cost pressure was likely to remain stable compared to the current level, in tandem with prices of oil and commodities in the world market. Nonetheless, strained factors of production, especially labor, could cause costs of production and inflationary pressure to accelerate in the periods ahead.

(3) Inflation expectations remained stable as reflected by the 12-month ahead inflation expectation in February 2013 of 3.5 percent which remained unchanged from the previous month **(Chart 2.21)**.

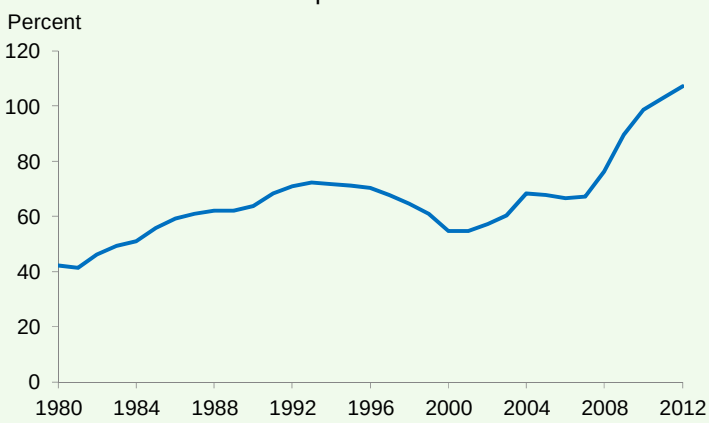
Risks to inflation that would warrant close monitoring going forward included the effects of the minimum wage hike and the impact of the energy price restructuring on inflation, which could be higher than assessed.

Budget sequestration and implications on the US economy

The origin of and the need for budget sequestration:

Budget deficit has been a chronic problem[1/] for the US economy. The need to borrow to fund the deficits have led to a persistent rise of public debt, which spiked sharply after the global financial crisis in 2008 when government stimulus measures were needed to shore up the economy **(Chart 1)**. This ongoing budget deficit problem has begun to affect long-term fiscal stability. With this concern, the US congress has become more cautious in approving debt ceiling increase in respond to the rise in sovereign debt. For the latest debt ceiling increase in 2011, the US congress enacted a Budget Control Act on August 2, 2011, which stipulated a condition for the US government to formulate a long-term plan to reduce the budget deficit by November 23, 2011 in exchange for a debt ceiling approval. Failure to do so would result in an automatic budget cut, so called "sequestration".

Chart 1 US public debt to GDP



Source: World Economic Outlook Database October 2012, IMF

The US government turned out to fail to meet the budget cut by the set deadline due to disagreements on revenue and expense measures of the two main political parties. Therefore, budget sequestration of 1.2 trillion US dollars was activated for 2013-2021, effective from March 1, 2013 onwards.[2/]

Details of budget sequestration

In FY 2013, the US government will need to reduce the overall budget by 85 billion US dollars, equivalent to 2.3 percent of the 2013 annual budget, or 0.6 percent of GDP. Most of the cuts would be on discretionary spending, including 43 billion US dollars on defense and 31 billion US dollars on other budgetary expenses. Meanwhile, cuts on mandatory spending of 11 billion US dollars will be targeted at Medicare expenses. However, other mandatory spending, such as social security and Medicaid will be exempted from the budget cuts.

[1/] Average revenue and expense to GDP between 1950-2008 registered at 18.0 and 19.8 percent, respectively. In 2012, the figures were approximately 15.8 and 24.0 percent, respectively.

[2/] Budget sequestration has been postponed from its former deadline on January 2, 2013 to March 1, 2013 as a result of a deal reached on the fiscal cliff problems.

Implications on the US economy

Budget sequestration would ultimately affect the US economy in 2013 but it would not cause the US economy to contract or enter into another recession due to the following reasons.

(1) The Congressional Budget Office (CBO) has assessed that even with a 85 billion US dollar budget cut, actual spending on the real economy that would be reduced would only amount to 42 billion US dollars because (i) some government departments and agencies would be able to disburse from the previous fiscal year's committed budget; and (ii) in general, some of the allocated budget would not be fully disbursed.

(2) Despite the abruptness of the budget cut, reduction in actual spending would be gradual. Government departments and agencies will gradually cut spending and postpone some expenses in a sequential manner. For example, new employment in the government sector would be postponed to be after a reduction in work hours for existing workers have been done. In this way, affected parties would have time to adjust and hence the impact of budget sequestration could be eased.

(3) There have been improvements in the US economic outlook whereby private demand, particularly consumption, business investment and real estate investment, have edged up. Such improvements would by and large help support the economy, compensating for lower government spending.

Risks in the periods ahead

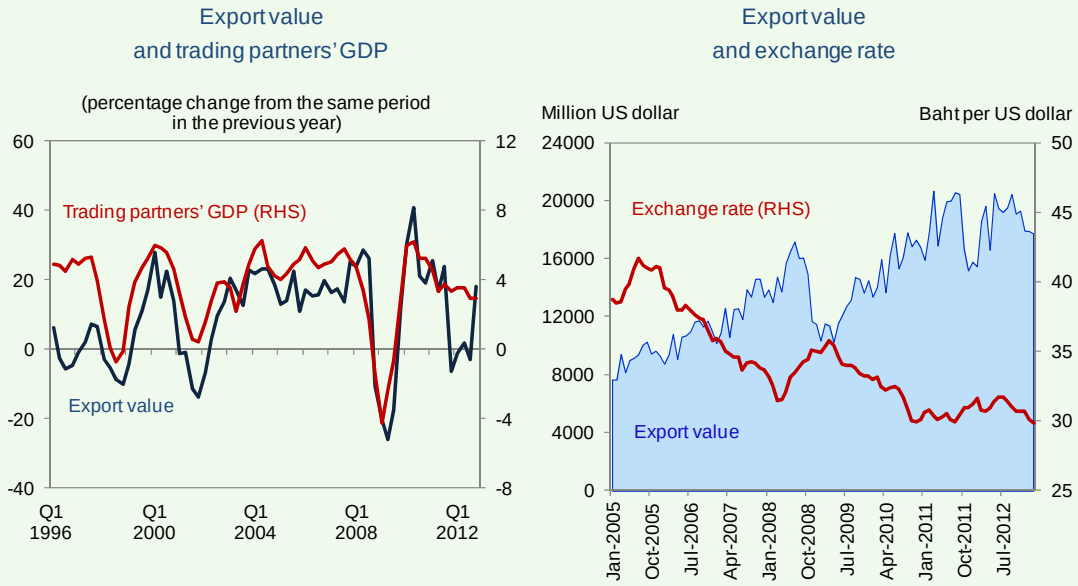
Fiscal problems remain the key risk to the US economy in the long run given that mandatory spending is on the rise in tandem with the aging population. Nevertheless, an immediate problem that the US government must deal with is the issue of debt ceiling that has been temporarily suspended by a Congressional ruling until May 19, 2013. Should the US government fail to find a solution to this problem, there is a risk that the economy would slow down further due to (1) a deterioration in private sectors' sentiment; (2) a potential downgrade by rating agencies due to government's default risk; and (3) another possible spending cut to reduce the deficit further in exchange for an increase in the debt ceiling as was the case in 2011.[3/]

In conclusion, though the budget sequestration would have some adverse effect on US economic growth but it would help improve the country's fiscal stability in the long-term. Nevertheless, risks to fiscal stability will remain with the US economy in the foreseeable future. The challenge now faced by the US government is to find a timely solution to its fiscal problems, as well as to strike the balance between short-term growth and long-term fiscal sustainability.

[3/] On August 2, 2011, US congress approved an increase in the debt ceiling of 2.1 trillion US dollars (from 14.3 trillion US dollars to 16.4 trillion US dollars) in exchange for a reduction in the deficit of 2.1 trillion US dollars within the next ten years.

Appreciation of the Thai baht and its implication on exports

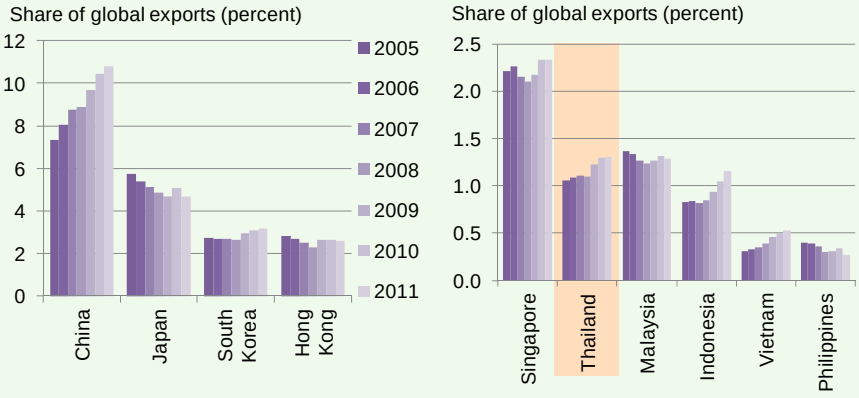
Chart 1 Thai export value compared with the exchange rate and GDP of trading partners



Sources: Customs Department and Bank of Thailand

The rapid appreciation of the Thai baht since the beginning of this year has led to concerns about its impact on Thailand's export sector which had been an important engine of growth for the Thai economy. However, our study reveals that Thai exports depend mainly on the economic condition of trading partners. As for a case in point, in 2010 when the Thai baht appreciated by 9.6 percent against the US dollar, Thai exports still grew strongly at 27.0 percent during that year **(Chart 1)**. Moreover, despite trend appreciation of the Thai baht, the share of Thai exports in total world exports has increased over time from 1.1 percent in 2005 to 1.3 percent in 2012 **(Chart 2)**, reflecting improving competitiveness of Thai businesses. However, even though the recent appreciation has not significantly affected overall export volume, the impact of baht strengthening on corporate income and profitability in Thai baht terms cannot be disregarded. Profits of Small and Medium Enterprises (SMEs) in particular, have been greatly affected.

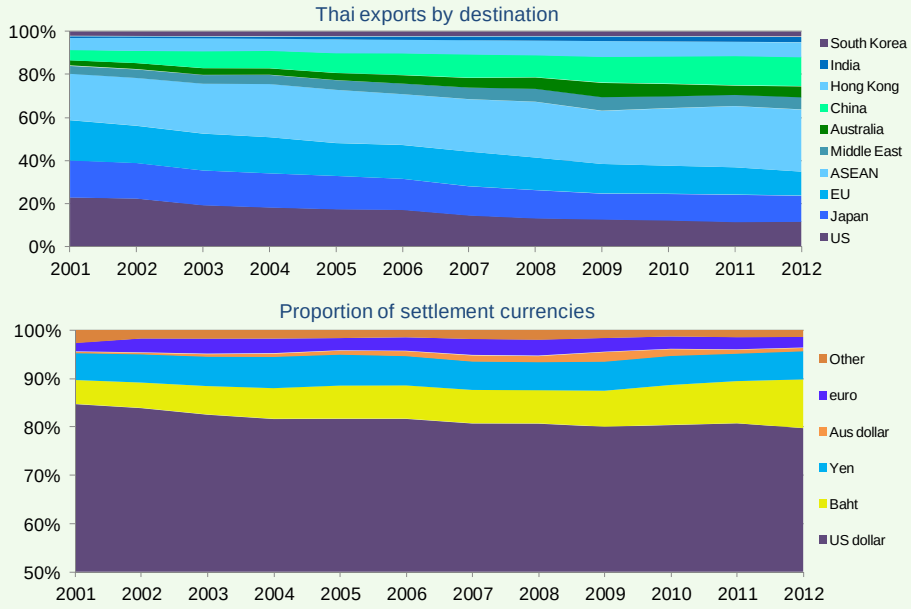
Chart 2 Market share by exporting country during 2005-2011



Source: International Trade Centre, Trademap and calculations by Bank of Thailand

In the past, Thai exporters have adjusted well to changing demand conditions and currency appreciation, as reflected by continued growth in exports amid the global economic crisis and strengthening of the baht. Businesses have diversified to new markets instead of just using pricing strategy to compete. There has been greater use of non-US dollar currencies for settlements. There has also been improvement in product sophistication of Thai export products which should help foster the exporters' bargaining and market power. Finally, businesses have also been hedging against exchange rate risks. All these adjustments have allowed Thai exporters to weather through the impact of exchange rate volatility. Details are as follows.

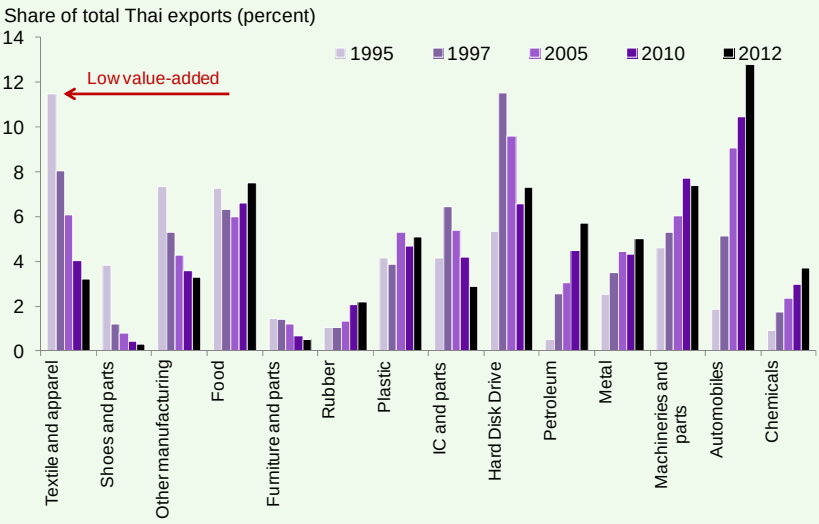
Chart 3 Share of Thai exports by destination and proportion of settlement currencies



(1) Seeking new markets and diversifying away from US dollar settlements: Exporters have diversified to new markets including ASEAN, India and the Middle East instead of relying only on major markets such as the US, the euro area and Japan where demand has been slowing. To reduce income volatility from exchange rate movements, exporters have also been diversifying settlement currencies instead of solely relying on the US dollar as in the past, with an increase in transactions denominated in Thai baht. This trend has been consistent with the expansion of exports to ASEAN markets, especially Cambodia, Laos, Myanmar and Vietnam **(Chart 3)**.

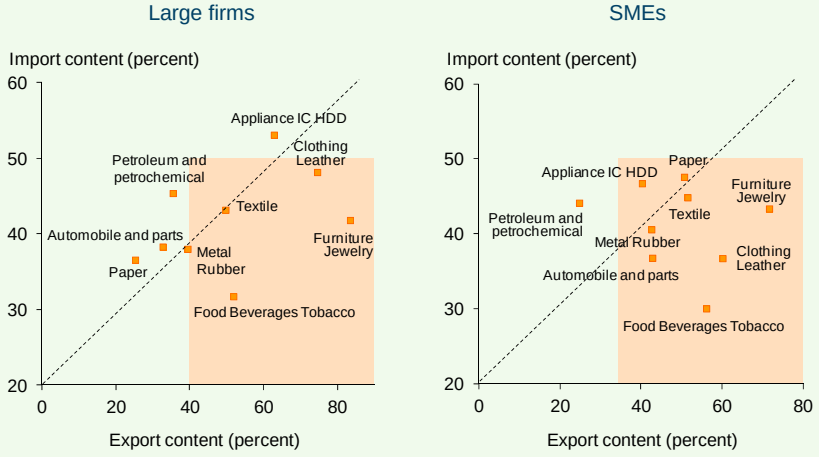
(2) Change in industrial structure: In recent years, Thailand has been shifting away from producing low value-added products such as textiles, shoes and food to more sophisticated products with higher value-added such as automobiles, chemicals and steel products **(Chart 4)**.

Chart 4 Changes in exporters' manufacturing structure



Source: Customs Department and Bank of Thailand

Chart 5 Exporters' natural hedge level



Note: Based on 1,310 large firms and 5,334 SME firms in the database.
Sources: Business Census 2007 National Statistics Office and calculations by Bank of Thailand

(3) Hedging against exchange rate risks by undertaking forward transactions in various currencies especially the US dollar**:** There has been a recent increase in hedging activities by exporters to protect themselves against the exchange rate risks. The proportion of forward transactions to total exports has increased from 24.8 percent in 2005 to 43. 8 percent in 2012.

(4) Resorting to natural hedge: Despite some adverse impact of the Thai baht appreciation on exports, exporters have also been able to reap benefits by importing capital and intermediate goods at lower costs. In particular, businesses with high import content such as those in automobile and petroleum industries would potentially be marginally affected by the Thai baht appreciation. On the other hand, businesses with low natural hedge such as furniture, jewelry and food industries would be more significantly affected as they mainly relied on domestic factors of production **(Chart 5)**.

In summary, although Thai exporters' income and profitability in Thai baht terms have been affected by the recent baht appreciation, on the other side of the coin this would provide an opportunity for businesses to import machineries and capital goods at lower costs. In this regard, productivity can be enhanced thereby helping businesses deal with other more structural problems of labor shortages and higher minimum wage. This will help improve long-term competitiveness of Thai businesses and sustainably support Thai exports in the future.

Monetary and Financial Stability

3. Monetary and Financial Stability

Thailand's overall economic and financial stability was well maintained. Improvement in global financial market sentiment improved, partly as a result of the agreement to avert the US fiscal cliff and positive US economic data, contributed to higher demand for bonds and equities in Thailand by both domestic and foreign investors. The Thai stock market exhibited higher volatility during this period owing to concerns over the Thai baht and uncertainties surrounding Cyprus' financial problems. Stability in other sectors, namely financial institutions, household, corporate, real estate and fiscal, remained sound.

Key risks to be monitored in the next periods included (1) the increase in retail investors' speculative activities; (2) the expansion in consumer loans which remained high, particularly auto leasings; (3) effects of the baht appreciation on corporate profitibility; and (4) impacts of the high growth of provincial real estate markets, rising construction costs, together with labor shortages, on the construction sector outlook and house prices.

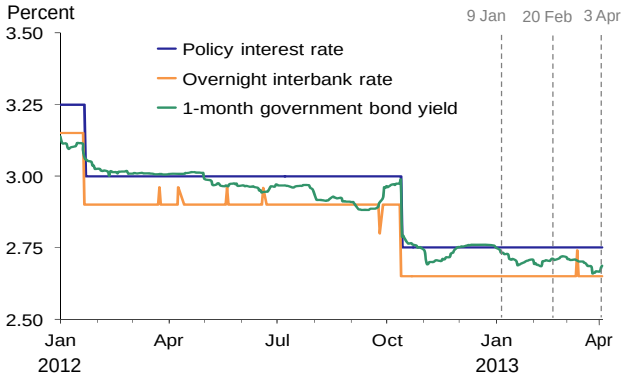
3.1 Financial markets

Money market rates remained close to the policy interest rate. Government bond yields were somewhat volatile due to demand factors and global financial market conditions. The stock market index rose over this period though the market became rather volatile towards the end of the quarter. Meanwhile, the Thai baht appreciated as a result of better financial market sentiment driven by improved global economic conditions and Thailand's strong fundamentals.

Money and bond markets

In 2013 Q1, short-term money market rates remained stable and close to the policy interest rate at 2.75 percent per annum as the MPC's decided to

Chart 3.1 Money market short-term interest rates

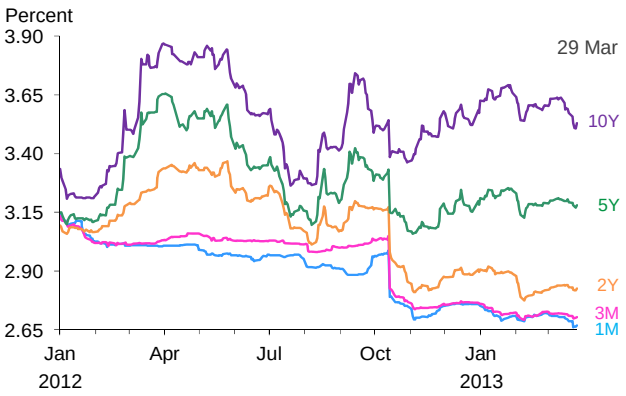


Sources: Bank of Thailand and the Thai Bond Market Association (ThaiBMA)

maintain the policy interest rate at this level at the January 9 and February 20 meetings. The overnight interbank lending rate and the one-month government bond yield stood at 2.65 and 2.67 percent, respectively, at the end of the quarter. At a subsequent meeting on April 3, 2013, the MPC continued to hold the policy interest rate at 2.75 percent per annum, causing short-term interest rates on April 3, 2013 to remain close to the 2013 Q1 end level **(Chart 3.1)**.

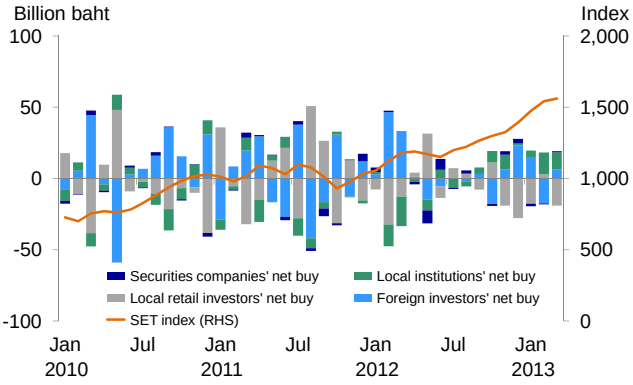
Short-term government bond yields in 2013 Q1 edged lower owing to an increase in demand by both domestic and foreign investors. Medium- to long-term government bond yields, however, increased in January thanks to improved global financial market sentiment partly as a result of the US fiscal cliff being averted and better-than-expected economic data, especially employment and real estate data. However, anticipation of a policy interest rate cut by some investors in February coupled with a continued rise in yields during the last 4-5 months resulted in a return in demand, which in turn caused bond yields to decline. Towards the end of the quarter, medium- to long-term government bond yields moved in tandem with the US government bond market and the gradual increase in foreign investors' holding of long-term government bonds **(Chart 3.2)**.

Chart 3.2 Government bond yields



Sources: The Thai Bond Market Association (ThaiBMA)

Chart 3.3 The Stock Exchange of Thailand index
(up to 29 March 2013)



Source: The Stock Exchange of Thailand

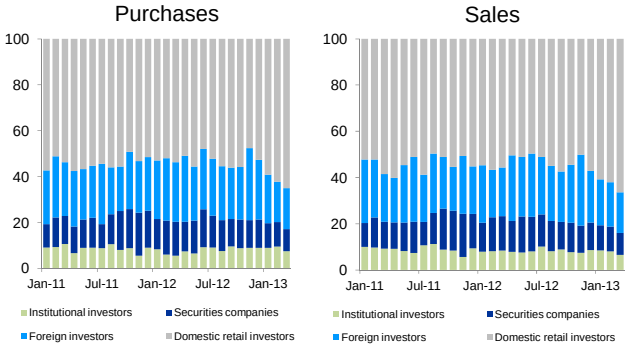
Equity market

The Stock Exchange of Thailand (SET) Index in 2013 Q1 was somewhat volatile, especially in March due to concerns over the Thai baht and uncertainties surrounding Cyprus' financial crisis. In this regard, the SET Index increased continuously since the beginning of the quarter in line with increasing foreign investors' and institutional investors' demand **(Chart 3.3)**. However, towards the end of March, the SET Index fell sharply by 119.16

points from a peak of 1,598.13 points on March 15, 2013 to stand at 1,478.97 points on March 22, 2013.

Key risks to be monitored in the periods ahead included the potential increase in retail investors' speculative activities since the beginning of the year. Signs of speculations were evident especially on small stocks, which could give rise to the risk of a stock market bubble **(Chart 3.4)**. To curb such risk, the Stock Exchange of Thailand (SET) and the Securities and Exchange Commission (SEC) announced a list of securities with abnormal trading patterns that were required to trade using cash balance account. Under this measure, investors must deposit cash in their trading account in full before they could start trading. The period required for cash trading for stocks on this cash basis list was also extended to six weeks from three earlier. The list of stocks that investors should take caution before trading would also be provided on the SET's website. Furthermore, cash collateral requirement for trading account was also increased to at least 20 percent from the previous requirement of 15 percent, effective from May 2, 2013 onwards.

Chart 3.4 Stock purchases and sales classified by types of investors

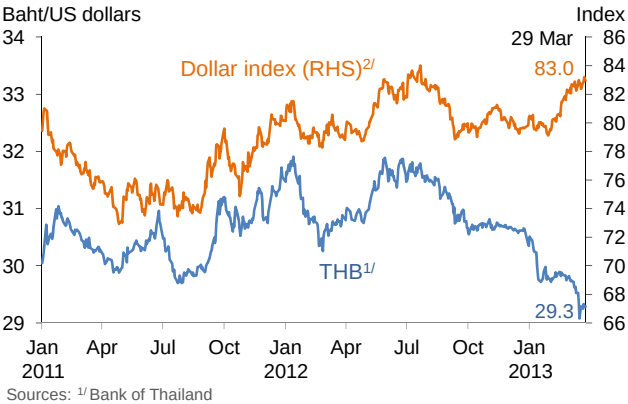


Source: Stock Exchange of Thailand

Foreign exchange market

At the end of 2013 Q1, the Thai baht stood at 29.29 baht per US dollar, appreciating by 4.31 percent from the end of the previous quarter **(Chart 3.5)** owing to both domestic and external factors. External factors arose from improved global financial market sentiment after (1) the agreement was reached to avoid the US fiscal cliff, (2) economic data for the US and China improved, and (3) advanced economies continued with aggressive monetary policy easing. Meanwhile, domestic factors included (1) the catching up effect of the Thai baht with regional currencies after some investors had previously postponed investments in Thailand due to concerns over post-flood recovery, (2) improved

Chart 3.5 Exchange rate and trade-weighted dollar index

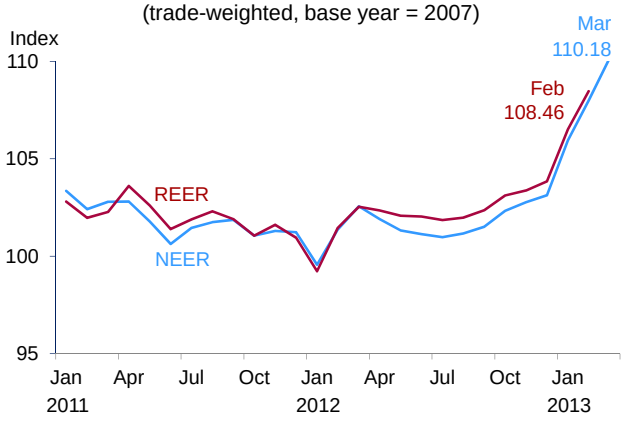


Sources: 1/ Bank of Thailand
2/ Bloomberg

outlook for the Thai economy as reflected by the upgrade of the country's credit rating by Fitch; (3) more concrete government investment plans; and (4) a gradual sale of US dollars associating with hedging activities of investors and exporters in anticipation of a stronger baht.

The above factors coupled with negative factors in many regional countries–for example, nuclear threat tension in the Korean Peninsula and impacts of the yen's depreciation on South Korea's economy, political uncertainties in Malaysia, and the persistent current account deficit in Indonesia–caused the baht to appreciate at a stronger pace compared to other regional currencies. As a result, the Nominal Effective Exchange Rate (NEER) in 2013 Q1 averaged at 108.04, appreciating by 5.16 percent from the previous quarter's average. Meanwhile, the Real Effective Exchange Rate (REER) also appreciated by 3.92 percent to average at 107.49[1/] in line with the NEER **(Chart 3.6)**.

Chart 3.6 NEER and REER
(trade-weighted, base year = 2007)



Source: Bank of Thailand

3.2 Financial institutions

In 2013 Q1, Commercial banks' deposit rates declined marginally from the end of 2012. Many commercial banks introduced special deposit products with higher interest rates to mobilize funds. Meanwhile, loan rates stabilized from the previous quarter. Private credits and deposits continued to register high growth.

[1/] Average over January - February 2013

Interest rates, credits and deposits

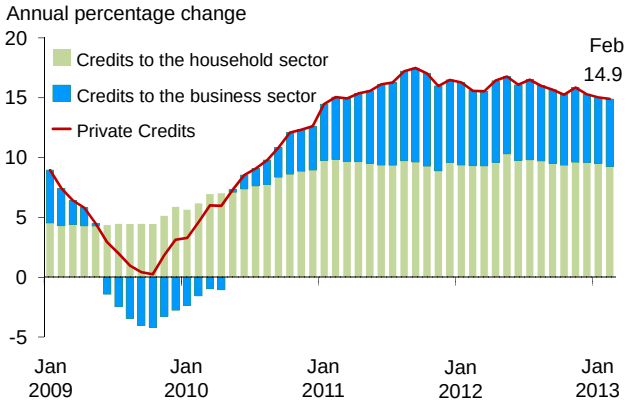
Interest rates on normal term deposits of large commercial banks[2/] in 2013 Q1 declined marginally from the end of 2012, partly due to adjustments of some deposit rates to be consistent within deposit brackets. At the end of 2013 Q1, interest rates on the 12-month deposits of the four largest commercial banks averaged at 2.40 percent per annum. However, during this quarter, many commercial banks also introduced special deposit products such as the 7-, 13- and 47-month time deposits which gave higher interest rates than the 12-month deposit rate in order to mobilize funds.

Meanwhile, the average MLR of the four largest commercial banks remained unchanged from end-2012 at 7.00 percent per annum.

Private credits at end-February 2013 continued to grow strongly at 14.9 percent from the same period last year, decelerating slightly, however, compared to the last quarter's growth of 15.3 percent **(Chart 3.7)**. Consumer loans continued to expand notably, especially auto leasings in tandem with the gradual delivery of cars under the government's first-car tax rebate scheme. Meanwhile, corporate loans also expanded well, for both investment and working capital in line with the investment upcycle.

Deposits and bills of exchange (B/E) at end-February 2013 continued to expand well, with an increase of 9.9 percent from the same period last year, although slowing somewhat from the previous quarter partly due to a continued contraction in B/E after the SEC announced a minimum B/E investment amount of 10 million baht, effective from July 1, 2012.

Chart 3.7 Other Depository Corporations' private credits



Source: Bank of Thailand

[2/] Bangkok Bank, Krungthai Bank, Kasikorn Bank and Siam Commercial Bank

This together with a slight slowdown in deposits results in a tightening of overall liquidity conditions.

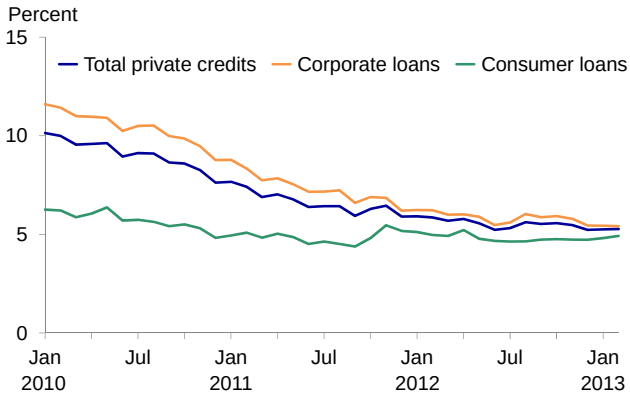
Stability of financial institutions

The overall stability of financial institutions in 2013 Q1 remained sound as reflected by the high capital adequacy ratio and low delinquency and NPL ratio. Meanwhile, commercial banks continued to record strong profitability.

Commercial banks continued to pose strong profitability whereby net profit for 2012 stood at 174.0 billion baht, increasing from 146.5 billion baht recorded in 2011. Moreover, the ratio of capital adequacy in January 2013 stood at 16.0 percent, declining slightly from 16.3 percent in December 2012 but remained significantly above the 8.5 percent legal requirement. With regard to loan quality in the banking system, despite some tightening in liquidity conditions as reflected by the loan to deposit and B/E ratio which stabilized at 93.1 percent **(Table 3.1)**, this was not yet a concern given that the delinquency and NPL ratio continued to stabilize at a low level of 5.3 percent **(Chart 3.8).**

Nevertheless, consumer loans continued to expand notably, in particular auto leasings owing to the first-car tax rebate scheme and personal loans that moved in line with higher spending. Moreover, commercial banks also shifted their focus to these types of lending given their high returns. These trends could potentially affect loan quality in the next periods.

Chart 3.8 Delinquency and NPL ratio



Source: Bank of Thailand

3.3 Non-financial sectors

Stability in non-financial sectors was well maintained. The recovery in both domestic and foreign economies coupled with public spending helped improve performance of the corporate sector. However, important risks that warranted monitoring in the next periods included accelerating consumer loans and household debt, especially auto leasings as well as the high expansion in real estate activities which when coupled with rising construction costs and labor shortages, could affect construction and cause housing prices to increase going forward.

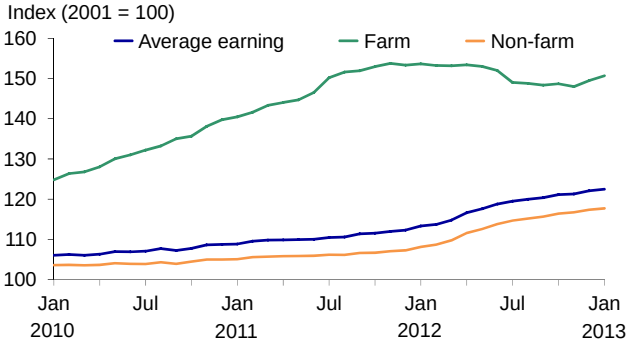
Household sector

Stability of the household sector remained sound thanks to high employment and household income. Nevertheless, households' debt servicing ability needed to be monitored closely given that consumer loans, especially auto leasings continued to exhibit rather high growth.

Households' income and employment remained healthy. Employment prospects were strong as reflected by the unemployment rate of only 0.7 percent in January 2013 as well as increases in average earnings for both farm and non-farm employment **(Chart 3.9)**. Debt servicing ability of households remained satisfactory as reflected by a low delinquency and NPL to consumer loans ratio of 4.7 percent.

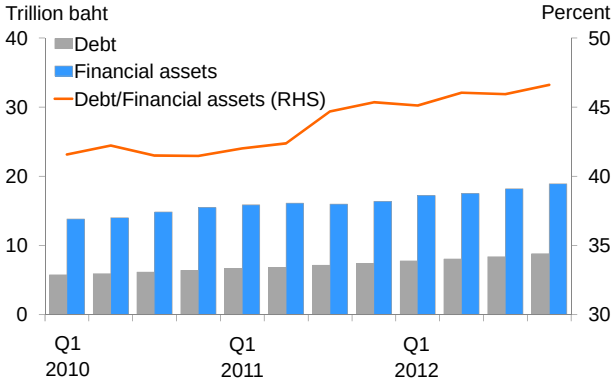
However, the persistent high growth of consumer loans led to some tightening in households' liquidity conditions as reflected by an increase in the ratio of debt to financial assets of households from 45.9 percent in 2012 Q3 to 46.6 percent in 2012 Q4 **(Chart 3.10)**. Moreover, there were some signs of increasing stress in

Chart 3.9 Households' income*



Note: *seasonally adjusted, real income (12-month moving average)
Sources: Office of Agricultural Economics, National Statistics Office, calculations by Bank of Thailand

Chart 3.10 Ratio of debt to financial assets



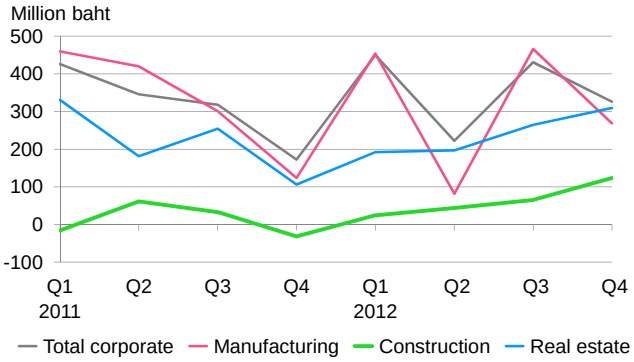
Source: Bank of Thailand

delinquencies of households with auto leasing. The increase in household debt which could potentially affect debt servicing ability will need to be monitored closely.

Corporate sector

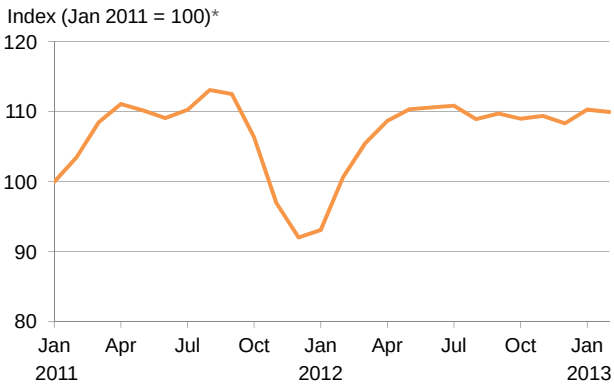
Corporate profitability in 2012 Q4 declined. Nevertheless, gradual export recovery in tandem with global economic recovery, continued domestic demand momentum, and fiscal spending were expected to help support improvements in corporate performance going forward.

Chart 3.11 Average net profit of non-financial companies listed on the stock exchange



Sources: Stock Exchange of Thailand and calculations by Bank of Thailand

Chart 3.12 Thailand's export value index



Note: *Seasonally adjusted, 3-month moving average
Sources: Customs Department, calculations by Bank of Thailand

Corporate profitability in 2012 Q4 declined, partly due to the impact of the slowing global economy on exports. The average net profit of firms in the manufacturing sector declined from 465.6 million baht in 2012 Q3 to 268.6 million baht in 2012 Q4 **(Chart 3.11)**, subsequently causing the operating profit margin (OPM) to decline from 5.1 percent to 3.8 percent. However, recovery in the domestic economy together with an increase in post-flood housing demand led to improvements in performance of related businesses from the previous quarter, particularly construction and real estate firms.

Looking ahead, it was expected that the gradual recovery in exports in line with global economic recovery, as well as continued domestic demand momentum and public spending would help support improvement in corporate performance **(Chart 3.12)**. Nonetheless, important risks that could affect stability of the corporate sector in the next periods included the impact of rising operating costs as a result of the minimum wage policy and labor shortages as well as the impact of the baht appreciation on corporate profit and liquidity, particularly for small and medium enterprises.

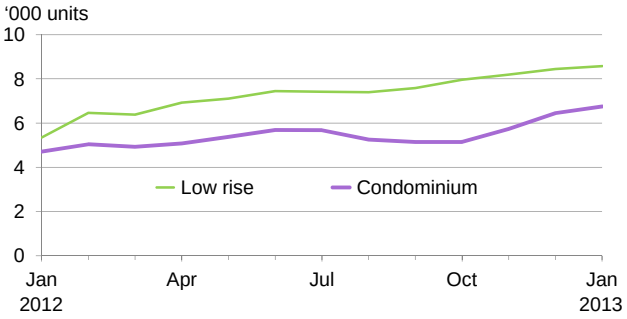
Real estate sector

The real estate sector in 2013 Q1 continued to expand well from the previous year, in particular condominiums. Stability in the real estate sector remained sound. However, important risks that warranted monitoring included the surge in provincial real estate activities, higher construction costs and labor shortages, which together could affect construction and cause housing prices to rise in the next periods.

During the start of 2013, the overall real estate sector continued to improve from the previous year, in particular condominiums with an increase in demand as reflected by a rise in the number of housing registrations in Bangkok and its vicinity in January 2013 compared to the previous month **(Chart 3.13).** Furthermore, a survey of real estate businesses also revealed that most of the demand was real demand, that is, for actual residency. On the supply side, it was found that real estate firms continued to launch new openings, especially in areas close to sky train extensions **(Chart 3.14)**.

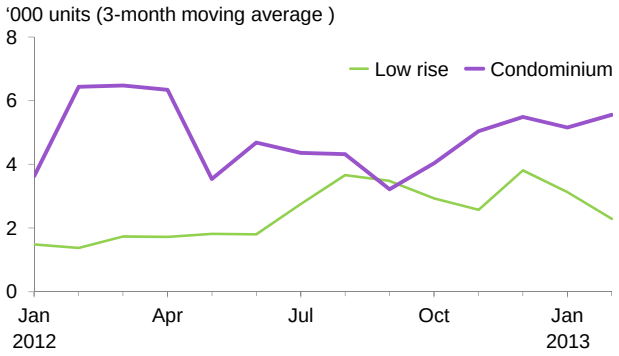
Concerning risks to the real estate sector, it was assessed that although housing prices increased, they were consistent with rising household income. This was reflected by the ratio of condominium price to income of home loan borrowers of 2.8, which remained close to the five-year average of 2.6. Moreover, commercial banks' home loan standard and debt servicing ability of home loan borrowers remained intact. Nevertheless, signs of a more relaxed lending standard were starting to be observed as reflected by the debt service coverage ratio (DSCR) in February 2013 of 4.4, which had been declining continuously since mid-2012 and was lower than its five-year average of 4.6 **(Chart 3.15)**.

Chart 3.13 Number of housing registrations in Bangkok and its vicinity*



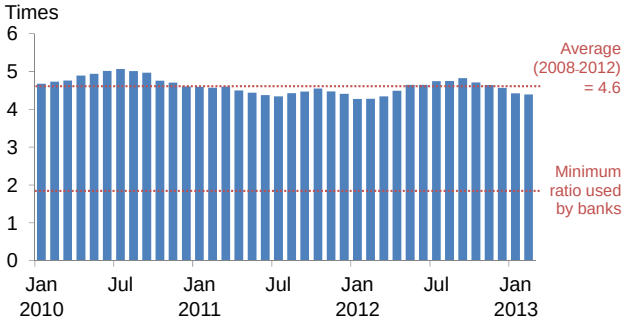
Note: *3-month moving average (data as of January 2013)
Number of housing registrations, including both new and pre-owned units, is used to reflect housing demand
Source: REIC

Chart 3.14 Number of new launches in Bangkok and its vicinity*



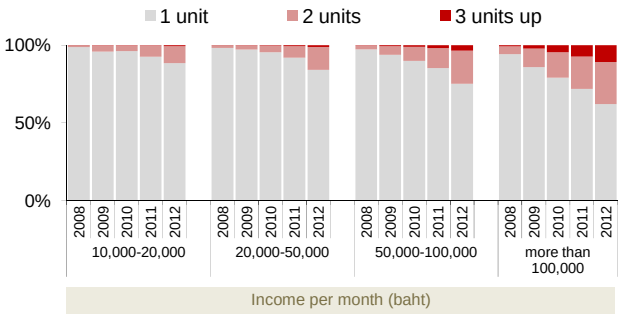
Note: *3-month moving average (seasonally adjusted data as of February 2013)
Source: Agency for Real Estate Affairs (AREA) and calculations by Bank of Thailand

Chart 3.15 Debt Service Coverage Ratio (DSCR)



Note:
(1) DSCR = monthly income/monthly debt payment (principal and interest)
(2) Calculated for new mortgages in Bangkok and its vicinity
(3) 50th percentile (3-month moving average)
Source: Agency for Real Estate Affairs (AREA) and calculations by Bank of Thailand

Chart 3.16 Volume of approved mortgages by number of housing units owned*



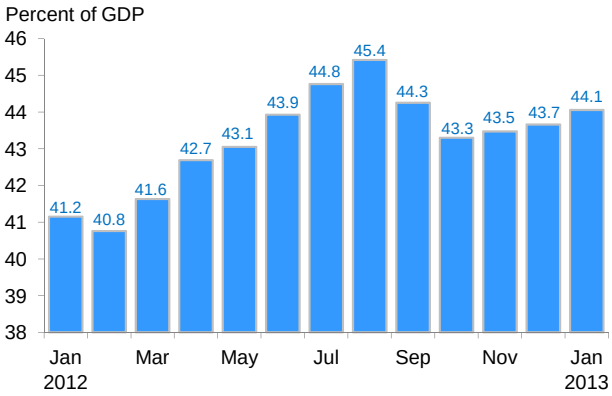
Note: *Data on mortgages for new and pre-owned homes from December 2007–December 2012
Source: Bank of Thailand

Although stability in the real estate sector remained sound overall, there were signs that people began to buy more than one house. This trend could pose risk to stability and would be monitored going forward **(Chart 3.16)**. In addition, a survey of provincial real estate businesses revealed that activities in large provinces, especially openings of condominiums, were on the rise. There were signs of speculative activities in some areas as reflected by a high level of reservations at new launches which led to rapid sold-outs as well as an increase in resale of reservation rights. Therefore, the expansion in such activities, the quality and standard of home loans as well as the impact of rising production costs and labor shortages would be monitored closely, as these factors could affect profitability of real estate firms, particularly smaller firms, and could result in higher housing prices, going forward.

Fiscal sector

Stability of the fiscal sector was well maintained but the ratio of public debt to GDP could rise going forward, owing to the increase in public spending.

Chart 3.17 Public debt to GDP



Note: Fiscal year
Source: Public Debt Management Office

Stability of the fiscal sector was well maintained as reflected by the ratio of public debt to GDP of 44.1 percent at end-January 2013, which was close to the previous quarter's level **(Chart 3.17)**. Looking ahead, however, the ratio of public debt to GDP could rise on the back of increasing public spending, especially from the rice pledging scheme, the Royal Decree on Investment Loans for Water Resource Management and Future Development as well as the transport infrastructure development loan bill. Therefore, fiscal burden could increase in the future.

Table 3.1 Sectoral Indicators for assessing risks and vulnerabilities to financial stability

Indicators	2009	2010	2011	2012				2013	
				Q1	Q2	Q3	Q4	Jan	Feb
1. Financial markets sector									
Bond market									
Bond spread (10 years–2 years)	2.1	1.3	0.4	0.3	0.5	0.3	0.6	0.8	0.8
Equity market									
SET Index	584.0	849.8	1,025.5	1,120.0	1,168.7	1,228.0	1,314.8	1,434.8	1,516.3
Actual volatility (SET Index)[1/]	26.0	17.6	21.3	13.1	16.7	11.0	8.9	9.5	11.5
Price to earnings ratio (times)	10.7	11.5	13.2	14.4	15.7	17.7	17.3	16.3	17.3
FX market									
Actual volatility (baht)	3.7	3.4	4.8	6.0	4.5	4.7	3.3	4.1	3.7
Nominal effective exchange rate (NEER)	98.5	102.9	101.8	101.2	101.4	101.2	102.7	106.0	108.0
Real effective exchange rate (REER)	102.0	102.9	97.5	101.1	102.1	102.1	103.4	106.6	n.a.
2. Financial institutions sector[2/]									
Minimum lending rate (MLR)[3/]	6.0	5.9	6.9	7.2	7.1	7.1	7.0	7.0	7.0
12-month fixed deposit rate[3/]	0.9	1.0	2.4	2.9	2.9	2.9	2.5	2.4	2.4
Capital adequacy									
Regulatory capital to risk-weighted asset (%)	16.1	16.2	15.1	15.2	15.1	15.9	16.3	16.0	n.a.
Earnings and profitability									
Net profit (billion baht)	92.2	123.0	147.2	40.1	49.1	45.2	39.3		
Return on assets (ROA)	0.9	1.1	1.2	1.2	1.4	1.3	1.2		
Liquidity									
Loan to deposit and B/E	85.8	88.3	89.8	90.3	92.4	91.4	93.1	93.1	n.a.
3. Household sector									
Debt to financial assets	41.3	41.5	45.4	45.2	46.1	45.9	46.6		
NPL and delinquency ratio (%)									
Financial institutions :									
Personal loans	6.1	6.0	5.8	5.6	5.8	5.8	5.8		
Credit cards	5.4	4.2	4.8	4.6	4.6	4.5	3.9		
Thai commercial banks :									
Consumer loans	6.2	4.8	5.2	4.9	4.7	4.7	4.7		
Mortgage loans	5.2	4.1	4.0	3.9	3.7	3.8	3.7		
Auto leasing	9.6	7.3	8.9	8.1	7.2	7.3	7.4		
4. Non-financial corporate sector [4/]									
Operating profit margin (%)	5.8	6.9	6.8	7.1	6.9	6.8	5.8		
Debt to equity ratio (times)	0.7	0.7	0.8	0.8	0.9	0.9	0.8		
Income coverage ratio (times)	3.2	4.7	4.1	3.6	3.6	3.5	3.6		
Current ratio (times)	1.5	1.5	1.5	1.5	1.4	1.4	1.5		

Table 3.1 Sectoral Indicators for assessing risks and vulnerabilities to financial stability (cont.)

Indicators	2009	2010	2011	2012				2013	
				Q1	Q2	Q3	Q4	Jan	Feb
5. Real estate sector									
3-month moving average of the number of approved mortgages from banks (Bangkok and its vicinity)	49,823	62,042	62,627	14,002	16,957	21,625	21,321	7,024	6,546
Single-detached and semi-detached houses	15,966	18,732	18,949	3,756	4,598	7,645	6,437	1,898	1,808
Townhouses and commercial buildings	16,269	21,020	21,756	5,022	6,057	7,324	7,183	2,432	2,436
Condominiums	17,588	22,290	21,922	5,224	6,302	6,656	7,701	2,694	2,302
3-month moving average of the number of new openings (Bangkok and its vicinity)	53,491	112,269	86,587	21,131	19,888	21,781	23,865	8,280	7,828
Single-detached and semi-detached houses	14,266	18,581	20,724	2,535	2,923	2,954	3,885	1,134	823
Townhouses and commercial buildings	18,846	33,734	25,002	2,051	2,404	6,941	5,425	1,992	1,454
Condominiums	22,379	59,954	40,861	16,545	14,561	11,886	14,555	5,154	5,551
Housing price index[5/]									
Single-detached houses (including land)	100.3	101.3	104.5	103.6	103.8	108.7	106.9	107.5	109.0
Townhouses (including land)	100.0	103.7	109.7	110.8	112.8	114.7	113.3	113.6	115.3
Condominiums	113.8	115.7	127.1	128.8	133.7	139.1	136.2	134.2	137.0
Land	108.2	113.0	121.7	129.9	122.2	125.7	129.7	134.8	135.7
6. Fiscal sector									
Public debt to GDP (%)	43.8	42.5	40.3	41.4	43.3	43.9	44.0	41.1	n.a.

[1/] Daily volatility (using exponentially weighted average method)

[2/] Based on data of all commercial banks

[3/] Average value of 4 largest Thai commercial banks

[4/] Only listed companies on SET (median)

[5/] Based on data of new approvals by commercial banks using hedonic regression method (January 2010 = 100)
(Due to the fact that the structure of the housing market has changed significantly, the Bank of Thailand is currently improving the price index to better reflect the structure change)